

Claude apps gateway on Azure: network-restricted deployment

Deployment, client connection and capacity planning for Claude Code, the VS Code extension and Claude Desktop with Claude models in Microsoft Foundry.

1. [Tutorial: Deploy the Claude apps gateway on Azure in a network-restricted environment](#)
2. [Connect Claude Code, VS Code and Claude Desktop to the gateway.](#)
3. [Plan capacity for 25,000 developers](#)

Tutorial: Deploy the Claude apps gateway on Azure in a network-restricted environment

A worked example that runs the Claude apps gateway in an internal Azure Container Apps environment, with Claude models in Microsoft Foundry behind a private endpoint, Azure Database for PostgreSQL with private access and Microsoft Entra ID sign-in. Each step is shown in the Azure portal, the Azure CLI and a script.

The Claude apps gateway is a server that ships inside Claude Code. It signs developers in through an OpenID Connect identity provider, applies per-group model access, and forwards Messages API requests to a model provider, so no model credential reaches a developer machine ([Claude apps gateway](#)). This tutorial is a worked example of the gateway on Azure, in the form of Anthropic's [AWS example](#): the gateway has no public endpoint, calls Claude in Microsoft Foundry through a private endpoint with its managed identity, and signs developers in with Microsoft Entra ID. Like the AWS example, it is a working example for customer-managed infrastructure, not a supported production deployment.

Each step is shown three ways:

Tab	Content
Azure portal	The portal path, the settings, and a screenshot of the resource this tutorial deployed on 2026-09-29
Azure CLI	The az commands, in PowerShell 7
Script	The step of <code>infra/azure-private/Deploy-Gateway.ps1</code> , which runs the same az commands (<code>infra/azure-private/Deploy-Gateway.ps1:5-20</code>)

The script reuses existing resources and may reconcile their settings, and `-Plan` prints the commands it would run without changing anything (`docs/adr/0005-network-restricted-deployment.md:56-60`).

In this tutorial, you:

- Create a virtual network, private DNS zones and a Foundry account with Claude deployments behind a private endpoint
- Create PostgreSQL with private access, a registry with the gateway image, and the gateway's managed identity
- Create an internal Container Apps environment, the Entra ID app registration and the gateway
- Check from a machine inside the network that every name resolves to a private address and the gateway is ready

Architecture

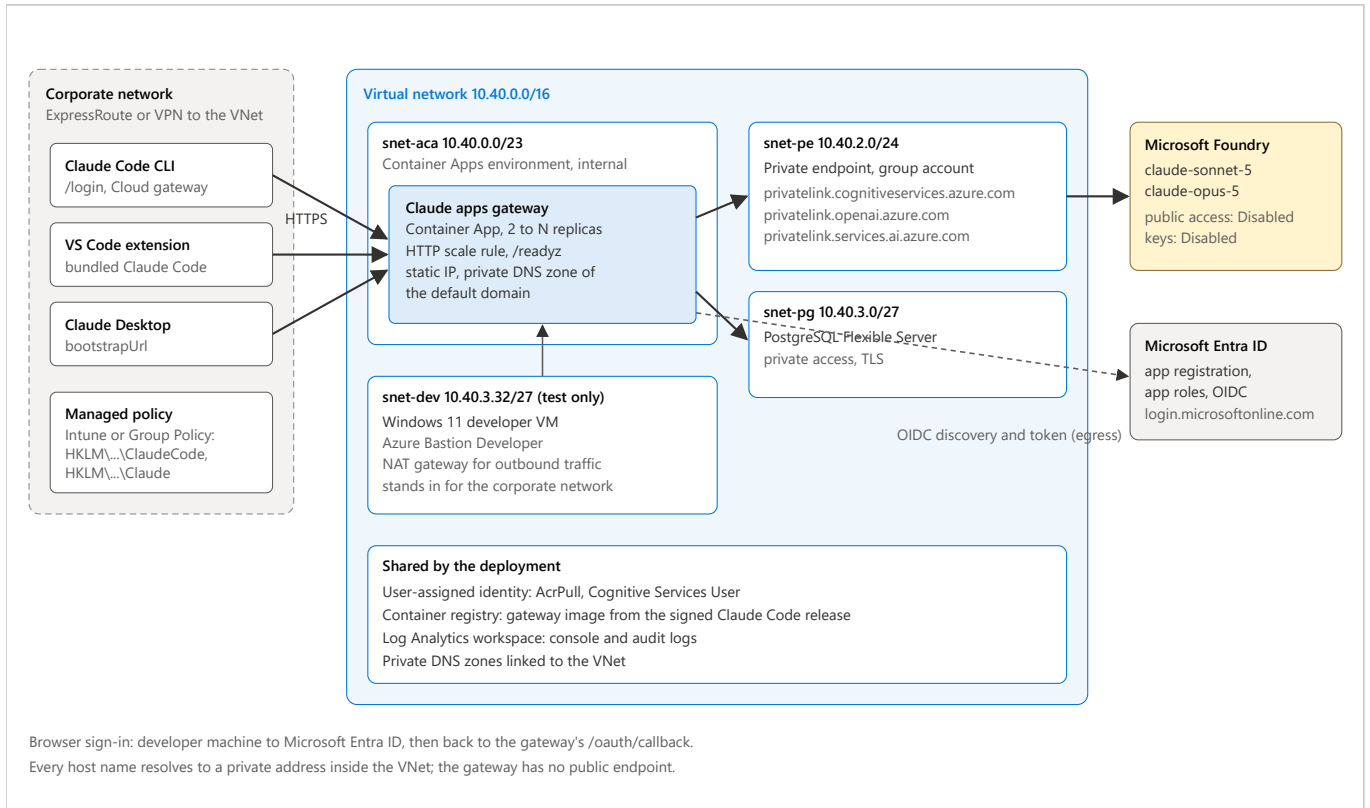


Diagram of the deployment: developer machines on the corporate network reach the gateway's private address over ExpressRoute or VPN; the gateway runs in an internal Container Apps environment in the snet-aca subnet, stores sign-in state in PostgreSQL in snet-pg, and calls Foundry through a private endpoint in snet-pe; browsers sign in to Microsoft Entra ID.

Developers reach the gateway at a host name that resolves to the environment's private static IP. /login in Claude Code accepts only a gateway whose host name resolves to private addresses ([prerequisites](#)). The deployment creates:

Resource	Name	Purpose
Virtual network	vnet-claude-gw, 10.40.0.0/16	Subnets snet-aca (/23, delegated to Microsoft.App/environments), snet-pe (/24, private endpoints), snet-pg (/27, delegated to Microsoft.DBforPostgreSQL/flexibleServers) and snet-dev (/27, test machine)
Container Apps environment	cae-claude-gw	Workload profiles, internal: a private static IP and no public endpoint
Container App	ca-claude-gw	The gateway: 2 to 3 replicas by default, with an HTTP scale rule
PostgreSQL flexible server	psql-claude-gw-<suffix>	The gateway's store (sign-in state and rate-limit counters), private access
Foundry account	ai-claude-gw-<suffix>	claude-sonnet-5 and claude-opus-5; public network access and key authentication disabled; private endpoint pe-foundry
Private DNS zones	Three Foundry zones, the environment's default domain, the PostgreSQL zone	Names resolve to private addresses inside the network
Container registry	acrclaudegw<suffix>	The gateway image, built from the signed Claude Code release
Managed identity	id-claude-gw	AcrPull on the registry and Cognitive Services User on Foundry
Log Analytics workspace	log-claude-gw	The gateway's console log, which carries its audit events
App registration	claude-apps-gateway-private-<suffix>	The OpenID Connect client, with app roles Gateway.Standard and Gateway.Premium
Test machine (optional)	vm-dev, bas-claude-gw, ng-dev	A Windows 11 VM reached through Azure Bastion Developer, standing in for a developer machine on the corporate network

The names come from the script (infra/azure-private/Deploy-Gateway.ps1:79-104). The test deployment runs in North Central US, with the Foundry account in East US 2, where the subscription holds Claude quota (docs/UNKNOWNNS.md:74). Production differences, such as a hub-and-spoke network and a custom domain, are in ADR-0005 (docs/adr/0005-network-restricted-deployment.md:64-73), and sizing for 25,000 developers is in [Plan capacity for 25,000 developers](#).

Prerequisites

Requirement	Detail	Source
Azure subscription	Owner, or Contributor with User Access Administrator: the steps create role assignments	https://learn.microsoft.com/en-us/azure/role-based-access-control/role-assignments-cli
Microsoft Entra ID	A role that registers applications and assigns app roles, such as Application Administrator or Cloud Application Administrator	https://learn.microsoft.com/en-us/entra/identity/enterprise-apps/assign-user-or-group-access-portal
Claude quota	Quota for each Claude model and deployment type in the Foundry account's region	https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/claude-models-quotas-limits
Claude terms	The organization's legal name, two-letter country code and industry, which each Claude deployment records to accept the Azure Marketplace offer	https://learn.microsoft.com/en-us/azure/developer/ai/how-to/deploy-claude-foundry#terms-of-use
One region for the regional resources	Capacity for a new Container Apps environment, PostgreSQL flexible server and, for the test machine, Bastion Developer	docs/UNKNOWNNS.md:74
Tools	Azure CLI with the containerapp extension, PowerShell 7.2 or later, and Node.js 22 or later for the configuration renderer	infra/azure-private/Deploy-Gateway.ps1:1
Companion files	The repository claude-apps-gateway-foundry: the script in infra/azure-private, the gateway configuration in config, the image definition in infra/azure-test/image and the Entra app manifest infra/azure-test/entra-app.json; the commands run from its root	infra/azure-private/Deploy-Gateway.ps1:95-98
Production network	ExpressRoute or VPN from the corporate network to the virtual network, and DNS that resolves the gateway's zone for developer machines	https://learn.microsoft.com/en-us/azure/dns/dns-private-resolver-overview

Get the files, sign in and set the variables

The companion files are in the public GitHub repository naveenneog/claude-apps-gateway-foundry, under the MIT license. The commands below clone it, and every later command runs from its root:

```
git clone https://github.com/naveenneog/claude-apps-gateway-foundry.git
Set-Location claude-apps-gateway-foundry
```

Sign in to the tenant, select the subscription, and add the Container Apps extension of the Azure CLI ([Container Apps CLI](#)):

```
az login --tenant "<tenant ID>"
az account set --subscription "<subscription ID>"
az extension add --name containerapp --upgrade
```

Every CLI command in this tutorial reads these variables, and later steps read variables that earlier steps set, in the same PowerShell session. Secrets and generated files go to a folder in the user's profile under %LOCALAPPDATA%, outside the repository:

```
$rg = 'rg-claude-gw-internal'
$location = 'northcentralus' # network, gateway, PostgreSQL and test machine
$foundryLocation = 'eastus2' # Foundry account, where the subscription holds Claude quota
$allowedDomain = 'contoso.com' # the domain of the email claim of the users the gateway admits
$subscriptionId = az account show --query id -o tsv
$tenantId = az account show --query tenantId -o tsv
```

```
# Six hexadecimal digits from the subscription and the resource group make the global names unique.
$hash = [Security.Cryptography.SHA256]::HashData([Text.Encoding]::UTF8.GetBytes("$subscriptionId/$rg"))
$suffix = [Convert]::ToHexString($hash).Substring(0, 6).ToLowerInvariant()
$vnet = 'vnet-claude-gw'
$foundry = "ai-claude-gw-$suffix"
$postgres = "psql-claude-gw-$suffix"
$acr = "acrclaudegw$suffix"
$identity = 'id-claude-gw'
$environment = 'cae-claude-gw'
$app = 'ca-claude-gw'
$work = Join-Path $env:LOCALAPPDATA "claude-gateway-deploy-$suffix-$(Get-Date -Format yyyyMMddHHmmss)"
New-Item -ItemType Directory -Path $work -ErrorAction Stop | Out-Null
# A new folder that only the operator's account may open: inherited permissions are removed, and the run stops
# unless exactly one entry, the operator's, remains.
icacls $work /inheritance:r /grant:r "${env:USERDOMAIN}\${env:USERNAME}:(OI)(CI)F" | Out-Null
if ($LASTEXITCODE -ne 0 -or @(Get-Acl $work).Access).Count -ne 1 { throw "The permissions of $work could not be
limited to $env:USERNAME." }
$work
```

The suffix is the one the script derives (infra/azure-private/Deploy-Gateway.ps1:70-71). Each run creates a new folder, whose name the last line prints; a later PowerShell session sets `$work` to that name again. `icacls` removes the inherited permissions and grants the operator's account full control, and a new folder has no other explicit entries, so other accounts have none ([icacIs](#)). The commands pass each secret to the Azure CLI as an `@<file>` argument, which the CLI reads from the file, so no secret is on a command line ([Azure CLI quoting](#); infra/azure-private/lib/Az.psm1:1-10). The secret files are written with `.NET's File.WriteAllText`, not a cmdlet, because PowerShell module logging records the parameters of cmdlets ([about Logging Windows](#); docs/adr/0004-operator-and-developer-tooling.md:106-112). The folder and its files stay after a failed or interrupted step until `Remove-Item -Recurse -Force $work` deletes them, as in [Clean up resources](#).

Each portal resource opens at `https://portal.azure.com/#@<tenant ID>/resource<resource ID>`; the screenshots in this tutorial were captured from those links.

Deploy the gateway

Step 1: Create the network

The environment's subnet is delegated to `Microsoft.App/environments` and is /27 or larger ([custom virtual network](#)); PostgreSQL's subnet is delegated to `Microsoft.DBforPostgreSQL/flexibleServers` ([private access](#)). The NAT gateway gives the test machine outbound access for installers and sign-in; production developer machines use the corporate network instead.

Azure portal

1. **Resource groups** > **Create**: name `ng-claude-gw-internal`, region **North Central US**.
2. **Virtual networks** > **Create**: name `vnet-claude-gw`; **IP addresses**: address space `10.40.0.0/16`.
3. For the test machine: **NAT gateways** > **Create**: name `ng-dev`; **Outbound IP**: a new public IP `pip-ng-dev` ([NAT gateway quickstart](#)).
4. **Virtual networks** > `vnet-claude-gw` > **Settings** > **Subnets** > **+ Subnet**, once for each subnet in the table after these steps.

Subnet	Range	Subnet delegation	NAT gateway
snet-aca	10.40.0.0/23	Microsoft.App/environments	None
snet-pe	10.40.2.0/24	None	None
snet-pg	10.40.3.0/27	Microsoft.DBforPostgreSQL/flexibleServers	None
snet-dev	10.40.3.32/27	None	ng-dev

Home > vnet-claude-gw

vnet-claude-gw | Subnets ☆ ...

Virtual network

Search

+ Subnet Refresh | Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

☐	Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	snet-aca	10.40.0.0/23	-	507	Microsoft...	-	-		
☐	snet-pe	10.40.2.0/24	-	248	-	-	-		
☐	snet-dev	10.40.3.32/27	-	27	-	-	-		
☐	snet-pg	10.40.3.0/27	-	27	Microsoft...	-	-		

Showing 4 subnets

Give feedback

Screenshot of the Subnets page of vnet-claude-gw with snet-aca 10.40.0.0/23 and snet-pg 10.40.3.0/27 delegated, snet-pe 10.40.2.0/24 and snet-dev 10.40.3.32/27.

Azure CLI

```
az group create --name $rg --location $location
az network vnet create -g $rg -n $vnet --location $location --address-prefixes 10.40.0.0/16
az network public-ip create -g $rg -n pip-ng-dev --location $location --sku Standard
az network nat gateway create -g $rg -n ng-dev --location $location --public-ip-addresses pip-ng-dev
az network vnet subnet create -g $rg --vnet-name $vnet -n snet-aca --address-prefixes 10.40.0.0/23 --delegations Microsoft.App/environments
az network vnet subnet create -g $rg --vnet-name $vnet -n snet-pe --address-prefixes 10.40.2.0/24
az network vnet subnet create -g $rg --vnet-name $vnet -n snet-pg --address-prefixes 10.40.3.0/27 --delegations Microsoft.DBforPostgreSQL/flexibleServers
az network vnet subnet create -g $rg --vnet-name $vnet -n snet-dev --address-prefixes 10.40.3.32/27 --nat-gateway ng-dev
```

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step network
```

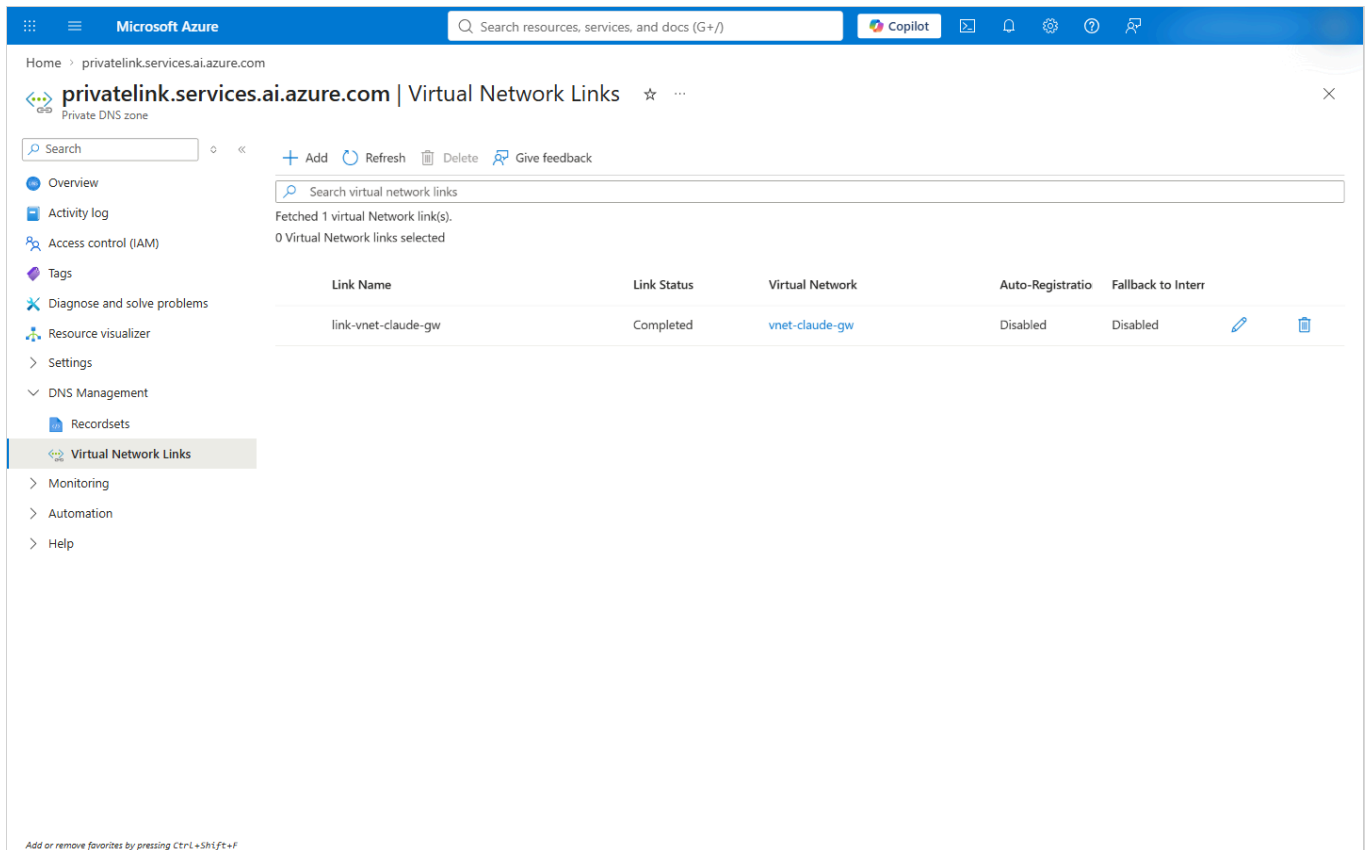
The step is in infra/azure-private/lib/Steps.Base.psm1:6-27.

Step 2: Create the private DNS zones for Foundry

A Foundry account answers on three host names, <account>.cognitiveservices.azure.com, <account>.openai.azure.com and <account>.services.ai.azure.com, and each family has its own private DNS zone ([private endpoint DNS](#)). Each zone is linked to the virtual network, so the names resolve to the private endpoint inside it.

Azure portal

1. **Private DNS zones > Create:** name privatelink.cognitiveservices.azure.com, resource group rg-claude-gw-internal.
2. The zone > **DNS Management > Virtual Network Links > Add:** name link-vnet-claude-gw, virtual network vnet-claude-gw, auto registration off ([private DNS quickstart](#)).
3. Repeat for privatelink.openai.azure.com and privatelink.services.ai.azure.com.



Screenshot of the Virtual Network Links page of `privatelink.services.ai.azure.com` with the link `link-vnet-claude-gw` to `vnet-claude-gw`, status `Completed`, auto registration `Disabled`.

Azure CLI

```
foreach ($zone in 'privatelink.cognitiveservices.azure.com', 'privatelink.openai.azure.com', 'privatelink.service
s.ai.azure.com') {
    az network private-dns zone create -g $rg -n $zone
    az network private-dns link vnet create -g $rg -z $zone -n "link-$vnet" -v $vnet -e false
}
```

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step dns
```

The step is in `infra/azure-private/lib/Steps.Base.psm1:30-43`.

Step 3: Create the Foundry account, the Claude deployments and the private endpoint

The account's public network access is disabled, so only the private endpoint reaches it, and local (key) authentication is disabled, so only Entra ID tokens are accepted ([disable local authentication](#)). Each Claude deployment carries a `modelProviderData` block with the organization's name, country code and industry, which accepts the Azure Marketplace offer for Claude ([terms of use](#)). The Azure CLI's `az cognitiveservices account deployment create` has no option for that block, so the CLI tab writes the deployment through Azure Resource Manager with `az rest` (`infra/azure-private/lib/Steps.Base.psm1:57-67`).

Azure portal

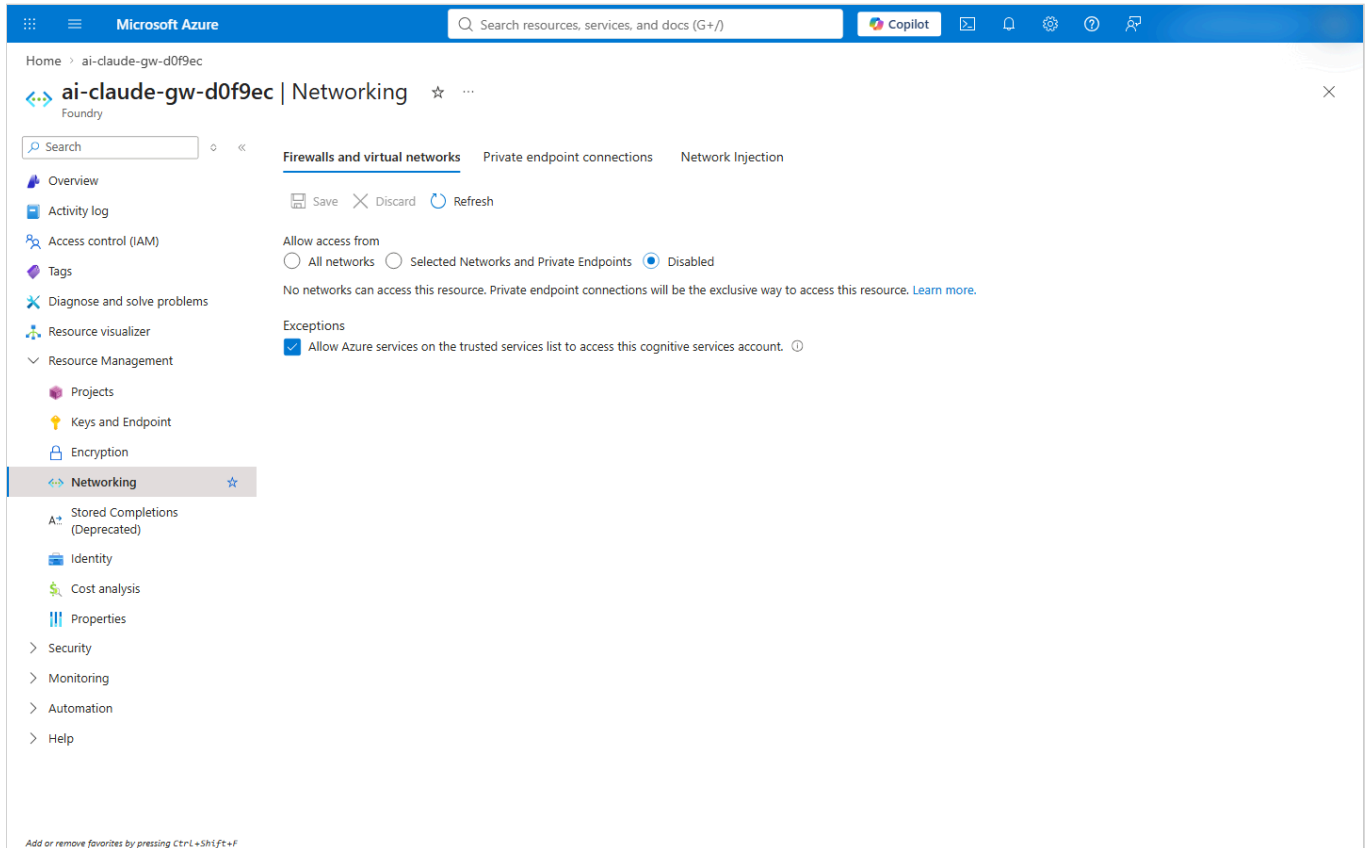
1. **Foundry** (Azure AI Foundry) > **Create** a resource: name `ai-claude-gw-<suffix>`, region **East US 2**.
2. **Foundry portal** (`ai.azure.com`), the resource > **Models + endpoints** > **Deploy model**: `claude-sonnet-5`, deployment type **Global Standard**; `claude-opus-5`, **Data Zone Standard**. The deployment dialog asks for the organization name, country and industry ([deploy Foundry models](#)).
3. The resource > **Resource Management** > **Networking** > **Private endpoint connections** > **+ Private endpoint**: name `pe-foundry`, region **North Central US**, target sub-resource account, virtual network `vnet-claude-gw` / `snet-pe`,

private DNS integration with the three zones of step 2 ([configure private link](#)).

4. **Networking > Firewalls and virtual networks > Allow access from: Disabled > Save.**

5. Key authentication: the [disable local authentication](#) guide sets `disableLocalAuth` with a template, PowerShell or Azure Policy; the CLI tab's `az resource update` sets it.

With public network access disabled, the Foundry portal manages the account only from a network with a path to its private endpoint ([configure private link](#)); the CLI tab creates the deployments through Azure Resource Manager, which does not need that path. The deployment recorded here created both deployments after public access was disabled.



Screenshot of the Networking page of the Foundry account with Allow access from set to Disabled: private endpoint connections are the only way to reach the resource.

Home > pe-foundry

pe-foundry | DNS configuration

Private endpoint

Search Add configuration Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Application security groups

DNS configuration

Properties

Locks

Monitoring

Automation

Help

Private DNS integration

To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint using a private DNS zone. You can also utilize your own DNS servers. [Learn more](#)

Customer Visible FQDNs

DNS records visible to the customer

Network Interface	IP addresses	FQDN
pe-foundry.nic.3a768ed6-61f4-4956-a20...	10.40.2.4	ai-claude-gw-d0f9ec.cognitiveservices.azure.com
	10.40.2.5	ai-claude-gw-d0f9ec.openai.azure.com
	10.40.2.6	ai-claude-gw-d0f9ec.services.ai.azure.com

Configuration name	FQDN	IP address	Subscription	Private DNS zone	DNS zone group
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Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

Screenshot of the DNS configuration of pe-foundry: the cognitiveservices, openai and services.ai host names of the account resolve to 10.40.2.4, 10.40.2.5 and 10.40.2.6.

Azure CLI

```
az cognitiveservices account create -g $rg -n $foundry --location $foundryLocation --kind AIServices --sku S0 --custom-domain $foundry --yes
$foundryId = az cognitiveservices account show -g $rg -n $foundry --query id -o tsv
az resource update --ids $foundryId --set properties.publicNetworkAccess=Disabled properties.disableLocalAuth=true

# One deployment per model; the terms block names your organization.
foreach ($m in @( @{ Name = 'claude-sonnet-5'; Sku = 'GlobalStandard' }, @{ Name = 'claude-opus-5'; Sku = 'DataZoneStandard' } )) {
    @{ sku = @{ name = $m.Sku; capacity = 10 }
      properties = @{ model = @{ format = 'Anthropic'; name = $m.Name; version = '2' }
                     modelProviderData = @{ organizationName = 'Contoso Ltd'; countryCode = 'US'; industry = 'technology' } } } |
    ConvertTo-Json -Depth 5 | Set-Content "deployment-$( $m.Name ).json"
    az rest --method put --url "https://management.azure.com/$foundryId/deployments/$( $m.Name )?api-version=2025-10-01-preview" --body "@deployment-$( $m.Name ).json"
}

az network private-endpoint create -g $rg -n pe-foundry --location $location --vnet-name $vnet --subnet snet-pe --private-connection-resource-id $foundryId --group-id account --connection-name foundry
az network private-endpoint dns-zone-group create -g $rg --endpoint-name pe-foundry -n foundry --private-dns-zone privatelink.cognitiveservices.azure.com --zone-name cognitiveservices
az network private-endpoint dns-zone-group add -g $rg --endpoint-name pe-foundry -n foundry --private-dns-zone privatelink.openai.azure.com --zone-name openai
az network private-endpoint dns-zone-group add -g $rg --endpoint-name pe-foundry -n foundry --private-dns-zone privatelink.services.ai.azure.com --zone-name services-ai
```

The capacity is in the quota units of the model and deployment type; the model versions are the ones this deployment used (infra/azure-private/Deploy-Gateway.ps1:87-91).

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step foundry -ClaudeOrganizationName 'Contoso Ltd' -ClaudeCount  
ryCode US -ClaudeIndustry technology
```

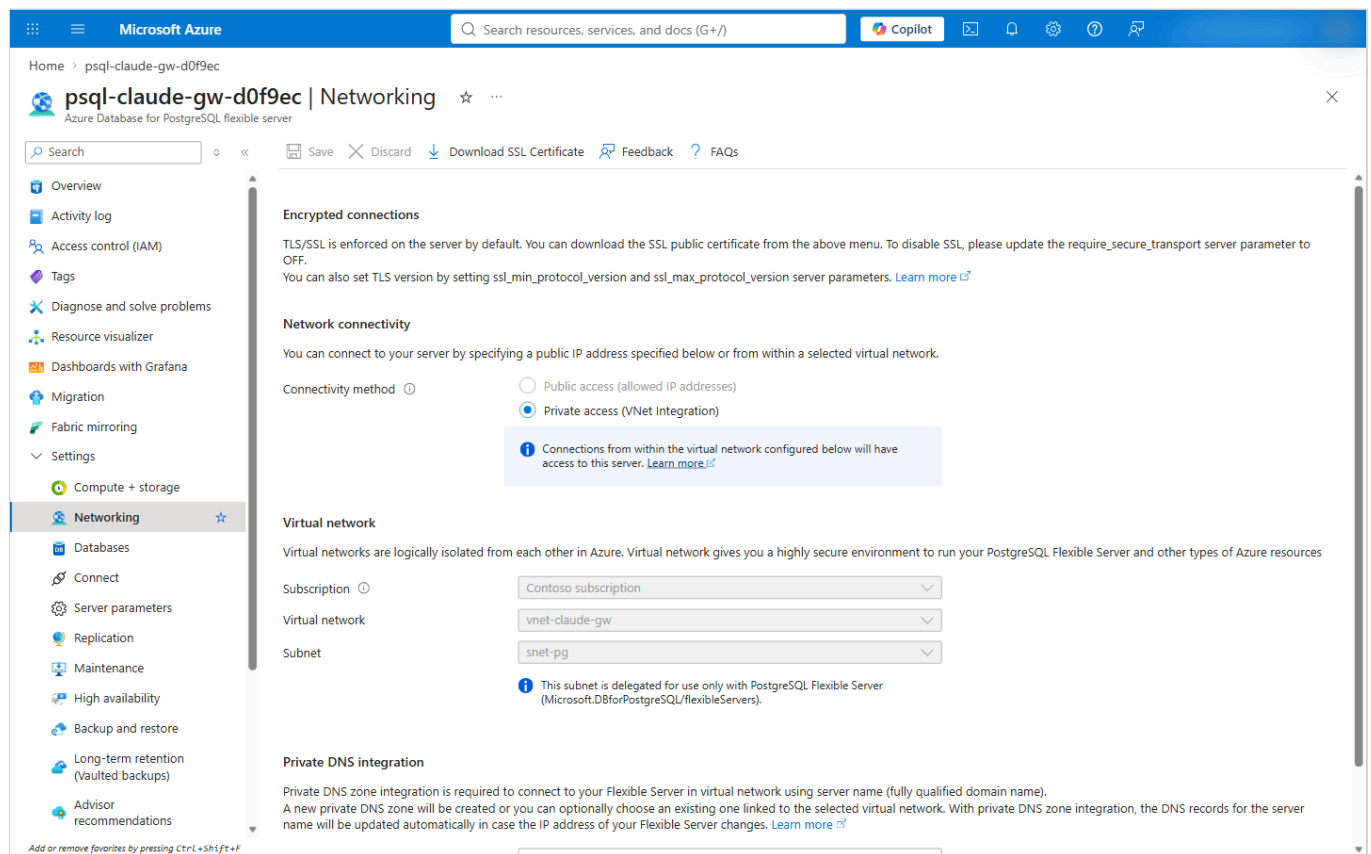
The step refuses to create a Claude deployment without `-ClaudeOrganizationName` (infra/azure-private/lib/Steps.Base.psm1:45-83).

Step 4: Create the PostgreSQL server

The gateway keeps device-flow grants and rate-limit counters in PostgreSQL 14 or later ([store](#)). The server uses private access: it has no public endpoint, and its name resolves through a private DNS zone linked to the virtual network ([private access](#)). Burstable B1ms is a test size with 35 user connections; sizing is in [Plan capacity for 25,000 developers](#).

Azure portal

1. **Azure Database for PostgreSQL flexible servers** > **Create**: name `psql-claude-gw-<suffix>`, region **North Central US**, PostgreSQL version **16**, **Compute + storage**: Burstable, Standard_B1ms, 32 GiB, high availability disabled; **Authentication**: PostgreSQL authentication, administrator `gatewayadmin` ([create a server](#)).
2. **Networking: Private access (VNet Integration)**, virtual network `vnet-claude-gw`, subnet `snet-pg`, a new private DNS zone `psql-claude-gw-<suffix>.private.postgres.database.azure.com`.
3. After creation: the server > **Settings** > **Databases** > **+ Add**: `gateway`.



Screenshot of the Networking page of the PostgreSQL server: connectivity method Private access (VNet Integration), virtual network `vnet-claude-gw`, subnet `snet-pg` delegated to PostgreSQL flexible servers.

Azure CLI

```
# A random password in a file; the CLI reads it from the file, and --output none keeps the reply, which repeats the  
# password, off the console.  
$pgPassword = ([Convert]::ToBase64String([Security.Cryptography.RandomNumberGenerator]::GetBytes(32)) -replace  
'[+/=]', '') + 'Aa1!'  
[IO.File]::WriteAllText("$work\pg-password.txt", $pgPassword)  
  
az postgres flexible-server create -g $rg -n $postgres --location $location --tier Burstable --sku-name Standard_B
```

```
1ms --version 16 --storage-size 32 --zonal-resiliency Disabled --vnet $vnet --subnet snet-pg --private-dns-zone
"$postgres.private.postgres.database.azure.com" --admin-user gatewayadmin --admin-password "@$work\pg-password.tx
t" --yes --output none
az postgres flexible-server db create -g $rg -s $postgres -d gateway
```

The reply of `az postgres flexible-server create` holds the administrator password and a connection string with it ([the command's source](#)). Step 9 reads `pg-password.txt`.

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step postgres -PostgresSku Standard_B1ms -PostgresTier Burstable
```

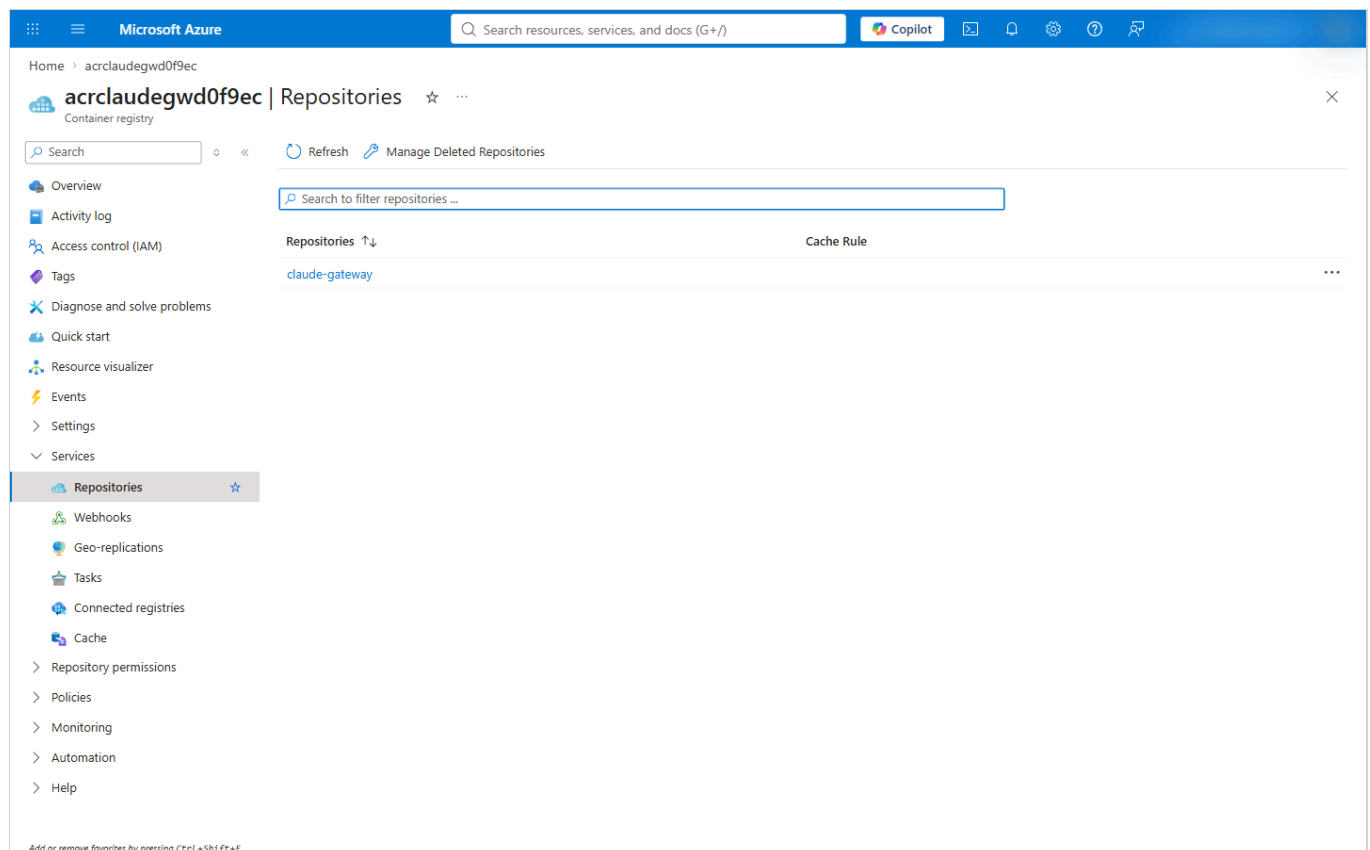
The script keeps the password in memory for the app step and sets a new one when the app step runs alone (`infra/azure-private/lib/Steps.App.psm1:94-100`).

Step 5: Create the registry and build the gateway image

The image runs the Claude Code binary as the gateway. Its build downloads the release, checks the release manifest's signature and the binary's SHA-256 checksum, and fails on a mismatch (`docs/adr/0005-network-restricted-deployment.md:71`; `infra/azure-test/image/Dockerfile:14-20`). `az acr build` runs the build as an ACR Tasks quick task in Azure ([ACR Tasks quickstart](#)). For a registry with a private endpoint and public access disabled, ACR Tasks runs the build on a dedicated agent pool in the virtual network ([agent pools](#)).

Azure portal

1. **Container registries** > **Create**: name `acrclauddegw<suffix>`, region **North Central US**, pricing plan **Basic**; admin user disabled ([registry quickstart](#)).
2. Build the image with the CLI tab's `az acr build`; the registry > **Services** > **Repositories** then lists `claud-gateway`.



Screenshot of the Repositories page of the container registry listing the claud-gateway repository.

Azure CLI

```
az acr create -g $rg -n $acr --location $location --sku Basic --admin-enabled false
az acr build --registry $acr --image claude-gateway:2.1.284 --build-arg CLAUDE_VERSION=2.1.284 --build-arg CLAUDE_SHA256=5cd90aabd83f8a15136c35aa37bb1d92b348993573316643dc3fe4e04afbf88f --no-logs infra/azure-test/image
$image = "$acr.azurecr.io/claude-gateway@" + (az acr repository show -n $acr --image claude-gateway:2.1.284 --query digest -o tsv)
```

--no-logs makes the CLI wait for the build and print the run as JSON, whose status is Succeeded for a good build; without it, the CLI prints the build log (tests/azure-private/fake-az.mjs:12-16). The checksum is the linux-x64 value in the release manifest of Claude Code 2.1.284 (infra/azure-private/Deploy-Gateway.ps1:93-94).

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step registry
```

The step stops when the build's status is not Succeeded (infra/azure-private/lib/Steps.Base.psm1:106-125).

Step 6: Create the gateway's identity

The gateway pulls its image and calls Foundry with a user-assigned managed identity. AcrPull lets it pull from the registry ([image pull with managed identity](#)), and Cognitive Services User lets it call the Foundry account; the gateway's Foundry upstream authenticates through DefaultAzureCredential ([Microsoft Foundry upstream](#)).

Azure portal

1. **Managed Identities** > **Create**: name id-claude-gw, region **North Central US** ([user-assigned identities](#)).
2. The registry > **Access control (IAM)** > **Add role assignment**: role **AcrPull**, assign access to **Managed identity**, member id-claude-gw ([assign roles in the portal](#)).
3. The Foundry account > **Access control (IAM)** > **Add role assignment**: role **Cognitive Services User**, member id-claude-gw.

Home > id-claude-gw

id-claude-gw | Azure role assignments ☆ ...

Managed Identity

Search ○ « + Add role assignment (Preview) Refresh

If this identity has role assignments that you don't have permission to read, they won't be shown in the list. [Learn more](#)

Subscription *
Contoso subscription

Role	Resource Name	Resource Type	Assigned To	Condition
AcrPull	acrclaudegwdf9ec	Container registry	id-claude-gw	None
Cognitive Services User	ai-claude-gw-d0f9ec	Foundry	id-claude-gw	None

Left sidebar: Overview, Activity log, Access control (IAM), Tags, **Azure role assignments**, Associated resources (preview), Resource visualizer, Settings, Monitoring, Automation, Help.

Footer: Add or remove favorites by pressing Ctrl+Shift+F

Screenshot of the Azure role assignments of id-claude-gw: AcrPull on the container registry and Cognitive Services User on the Foundry account.

Azure CLI

```

az identity create -g $rg -n $identity --location $location
$principalId = az identity show -g $rg -n $identity --query principalId -o tsv
$identityClientId = az identity show -g $rg -n $identity --query clientId -o tsv
$identityId = az identity show -g $rg -n $identity --query id -o tsv
$acrId = az acr show -g $rg -n $acr --query id -o tsv
az role assignment create --assignee-object-id $principalId --assignee-principal-type ServicePrincipal --role AcrP
ull --scope $acrId
az role assignment create --assignee-object-id $principalId --assignee-principal-type ServicePrincipal --role "Cog
nitive Services User" --scope $foundryId

```

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step identity
```

The step is in `infra/azure-private/lib/Steps.Base.psm1:127-145`.

Step 7: Create the internal Container Apps environment

An internal environment has no public endpoint: its apps are reachable from the virtual network at the environment's static IP ([custom virtual network](#)). A private DNS zone named after the environment's default domain, with a * record for the static IP, resolves the app's host name inside the network ([private DNS zone for an environment](#)).

Azure portal

1. **Log Analytics workspaces** > **Create**: name `log-claude-gw`, region **North Central US**.
2. **Container Apps Environments** > **Create**: name `cae-claude-gw`, region **North Central US**, environment type **Workload profiles**; **Monitoring**: Log Analytics `log-claude-gw`; **Networking**: use your own virtual network **Yes**, `vnet-claude-gw`, infrastructure subnet `snet-aca`, **Virtual IP**: **Internal**.
3. The environment > **Overview**: note the **Static IP**; the **Default domain** is in **JSON View** as `properties.defaultDomain`.
4. **Private DNS zones** > **Create**: the default domain as the zone name. The zone > **DNS Management** > **Recordsets** > + **Add**: name `*`, type `A`, the static IP. **Virtual Network Links** > **Add**: `vnet-claude-gw`.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and various icons. The main content area displays the 'cae-claude-gw' Container Apps Environment overview. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Apps, Services, Diagnostic tools (Preview), Monitoring, Automation, and Help. The main content area is divided into sections: Essentials, Applications, Get started, Monitoring, and Tutorials. The Essentials section provides key details about the environment, including Resource group, Status, Location, Subscription, and various configuration options. The Applications section shows a table of applications running in the environment.

Essentials	
Resource group (move)	: rg-claude-gw-internal
Status	: Succeeded
Location (move)	: North Central US
Subscription (move)	: Contoso subscription
Subscription ID	: 00000000-0000-0000-0000-000000000000
Aspire Dashboard	: Not yet active (set up)

Applications	
Name ↑	App Type
ca-claude-gw	Container App

Screenshot of the overview of `cae-claude-gw`: environment type Workload profiles, virtual network `vnet-claude-gw`, infrastructure subnet `snet-aca`, static IP 10.40.1.231, one application `ca-claude-gw`.

The screenshot shows the Azure portal interface for the resource **cae-claude-gw** under the **Container Apps Environment**. The **Networking** tab is selected in the left-hand navigation pane. The main content area shows the following settings:

- General**: Overview, Ingress settings, Request Routing, Encryption, Custom DNS Suffix.
- Public Network Access**: A toggle switch is set to **Disable**. A message states: "These settings are currently disabled since your app environment is internal."
- Virtual network**: A message states: "These options can't be modified post create." Below this, the configuration is shown:

Virtual network	vnet-claude-gw
Subnet	snet-aca
Virtual IP	Internal
Infrastructure resource group	rg-claude-gw-internal
- Private Endpoints**: A message states: "Private endpoints enable secure inbound access from the chosen virtual network. When configured, your app cannot be configured to allow public network access. [Get more info](#)". Below this, it shows "0 Private Endpoints".

At the bottom of the settings pane, there are **Apply** and **Discard** buttons.

Screenshot of the Networking page of cae-claude-gw: public network access disabled because the environment is internal, virtual network vnet-claude-gw, subnet snet-aca, virtual IP Internal.

The screenshot shows the Azure portal interface for the **Recordsets** page of the private DNS zone **politebush-4e216865.northcentralus.azurecontainerapps.io**. The **Recordsets** tab is selected in the left-hand navigation pane. The main content area shows the following details:

- Overview**: A record set is a collection of records in a zone that have the same name and are the same type. Record Sets will be automatically fetched in batches of 100 as you scroll through the existing record sets. [Learn more](#)
- Search**: A search bar is present.
- Recordsets**: A table displays the fetched record sets.

Name	Type	TTL	Value	Auto registered
*	A	3600	10.40.1.231	False
@	SOA	3600	Email: azureprivatedns-host.microsoft.com Host: azureprivatedns.net Refresh: 3600 Retry: 300 Expire: 2419200 Minimum TTL: 10 Serial number: 1	False

At the bottom of the recordsets pane, there are **Apply** and **Discard** buttons.

Screenshot of the record sets of the private DNS zone named after the environment's default domain, with a wildcard A record for the environment's static IP.

Azure CLI

```
az monitor log-analytics workspace create -g $rg -n log-claude-gw --location $location --retention-time 30
$workspaceId = az monitor log-analytics workspace show -g $rg -n log-claude-gw --query customerId -o tsv
```

```
$workspaceKey = az monitor log-analytics workspace get-shared-keys -g $rg -n log-claude-gw --query primarySharedKey -o tsv
[IO.File]::WriteAllText("$work\workspace-key.txt", $workspaceKey)
$aSubnetId = az network vnet subnet show -g $rg --vnet-name $vnet -n snet-aca --query id -o tsv
az containerapp env create -g $rg -n $environment --location $location --enable-workload-profiles true --infrastructure-subnet-resource-id $aSubnetId --internal-only true --logs-destination log-analytics --logs-workspace-id $workspaceId --logs-workspace-key "@$work\workspace-key.txt"

$domain = az containerapp env show -g $rg -n $environment --query properties.defaultDomain -o tsv
$staticIp = az containerapp env show -g $rg -n $environment --query properties.staticIp -o tsv
az network private-dns zone create -g $rg -n $domain
az network private-dns link vnet create -g $rg -z $domain -n "link-$vnet" -v $vnet -e false
az network private-dns record-set a add-record -g $rg -z $domain -n "*" -a $staticIp
```

The environment in this tutorial got the default domain `politebush-4e216865.northcentralus.azurecontainerapps.io` and the static IP `10.40.1.231`.

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step environment
```

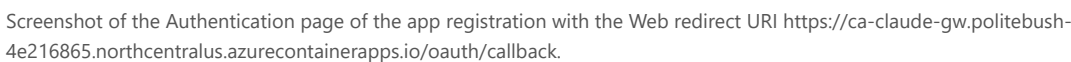
The step is in `infra/azure-private/lib/Steps.App.psm1:5-38`.

Step 8: Register the gateway in Microsoft Entra ID

The gateway is a confidential OpenID Connect client. Its redirect URI is `<public URL>/oauth/callback`, Entra ID issues v2.0 tokens, and the gateway admits a user whose token carries one of its app roles in the `roles` claim ([identity provider setup](#); `config/gateway.azure-private.yaml:22-23`). The app roles, the `email` optional claim and the delegated permissions `openid`, `profile`, `email` and `offline_access` come from `infra/azure-test/entra-app.json`.

Azure portal

1. **Microsoft Entra ID > App registrations > New registration:** name `claude-apps-gateway-private-<suffix>`, single tenant, redirect URI **Web** `https://ca-claude-gw.<default domain>/oauth/callback` ([register an application](#)).
2. The app > **App roles > Create app role:** `Gateway.Standard` and `Gateway.Premium`, allowed member types **Users/Groups** ([app roles](#)).
3. **Token configuration > Add optional claim:** ID token, email.
4. **Manifest:** `api.requestedAccessTokenVersion = 2` ([app manifest](#)).
5. **Certificates & secrets > New client secret:** **Expires** set to a custom 7 days, or the period the organization's credential policy sets; the gateway reads the secret as `OIDC_CLIENT_SECRET` in step 9 ([add a client secret](#)).
6. **Enterprise applications > the app > Users and groups > Add user/group:** a user or group, with a role ([assign users and groups](#)).



```

ConvertTo-Json -Depth 10 -InputObject $manifest.optionalClaims | Set-Content "$work\optional-claims.json"
ConvertTo-Json -Depth 10 -InputObject $manifest.requiredResourceAccess | Set-Content "$work\graph-access.json"

$appId = az ad app create --display-name "claude-apps-gateway-private-$suffix" --sign-in-audience AzureADMyOrg --w
eb-redirect-uris "https://$app.$domain/oauth/callback" --app-roles "@$work\app-roles.json" --optional-claims "@$wo
rk\optional-claims.json" --required-resource-accesses "@$work\graph-access.json" --query appId -o tsv
az ad app update --id $appId --set api.requestedAccessTokenVersion=2
$spId = az ad sp create --id $appId --query id -o tsv

# Give yourself Gateway.Premium; other users and groups get a role the same way, or in the portal.
$premium = ($manifest.appRoles | Where-Object value -eq 'Gateway.Premium').id
$me = az ad signed-in-user show --query id -o tsv
@{ principalId = $me; resourceId = $spId; appRoleId = $premium } | ConvertTo-Json | Set-Content "$work\role-assign
ment.json"
az rest --method POST --url "https://graph.microsoft.com/v1.0/servicePrincipals/$spId/appRoleAssignedTo" --body "@
$work\role-assignment.json" --headers Content-Type=application/json

```

A new service principal can take a few seconds to reach Microsoft Graph; the script retries the role assignment (infra/azure-private/lib/Steps.App.psm1:69-73).

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step entra -OperatorRole Gateway.Premium
```

The script creates the client secret in the app step, valid for -ClientSecretDays days (default 7). With -RotateClientSecret it adds a new secret and restarts the latest revision, because a changed secret reaches a revision only when the revision restarts or a new one is deployed ([manage secrets](#); infra/azure-private/lib/Steps.App.psm1:118-121). The old secret stays valid until it expires, since --append keeps it.

Step 9: Configure and deploy the gateway

The gateway reads gateway.yaml, rendered from the admin file config/gateway-admin.azure-private.json: the app roles it admits, the models and their Foundry deployments, a policy per role with its availableModels, the networks developers connect from, and the sign-in limits (scripts/admin/new-gateway-config.mjs:196-236). Values specific to the deployment and every secret are \${VAR} references that the Container App sets as environment variables ([secret expansion](#)).

The configuration's network settings for this topology:

```

listen:
  public_url: "${GATEWAY_PUBLIC_URL}"
  trusted_proxies: [100.100.0.0/17, 100.100.128.0/19, 100.100.160.0/19, 100.100.192.0/19] # where the ingress co
nnects from
store:
  max_connections: 5
  readiness_grace_seconds: 300 # keeps /readyz ready through a database failover
upstreams:
  - provider: foundry
    resource: "${FOUNDRY_RESOURCE}"
    auth: { use_azure_ad: true }
access_control:
  allow_cidrs: [10.40.0.0/16] # deployment.developerCidrs in the admin file

```

The full file is config/gateway.azure-private.yaml:8-73. trusted_proxies names the ranges a workload profile environment reserves, where the Container Apps ingress connects from; the ingress puts the client's address rightmost in X-Forwarded-For, so the gateway sees each developer's own address, and a developer cannot set X-Forwarded-For to take another address's sign-in limit (docs/adr/0005-network-restricted-deployment.md:75-88). The gateway container's secrets and settings:

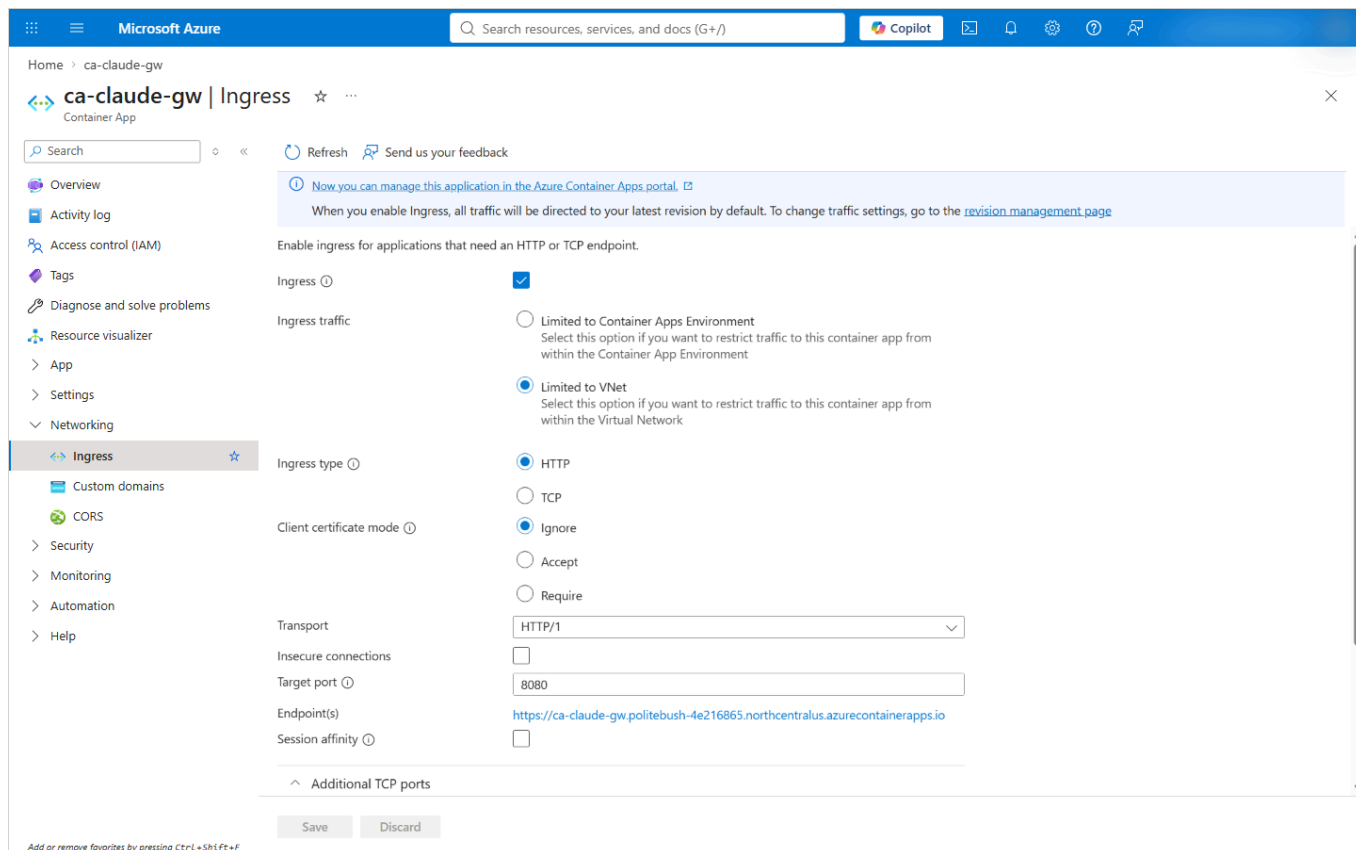
Setting	Value	Portal location
Image	acrclaudegw<suffix>.azurecr.io/claude-gateway@<digest>, pulled with id-claude-gw	Containers > Edit and deploy

Setting	Value	Portal location
Command and arguments	/usr/local/bin/claude with gateway --config /etc/claude/gateway.yaml	Containers > the container
CPU and memory	1 vCPU, 2 GiB	Containers > the container
Secrets	gateway-config (the rendered YAML), oidc-client-secret, jwt-secret, pg-password	Security > Secrets
Volume	Secret volume gateway-config with the file gateway.yaml, mounted at /etc/claude	Containers > Volumes
Environment variables	Listed in the CLI tab	Containers > the container > Environment variables
Health probes	Startup and liveness on /healthz, readiness on /readyz, port 8080	Containers > the container > Health probes
Ingress	Enabled, Limited to VNet , HTTP, target port 8080	Networking > Ingress
Scale	2 to 3 replicas, HTTP rule of 150 concurrent requests	App > Scale
Identity	User assigned: id-claude-gw	Security > Identity

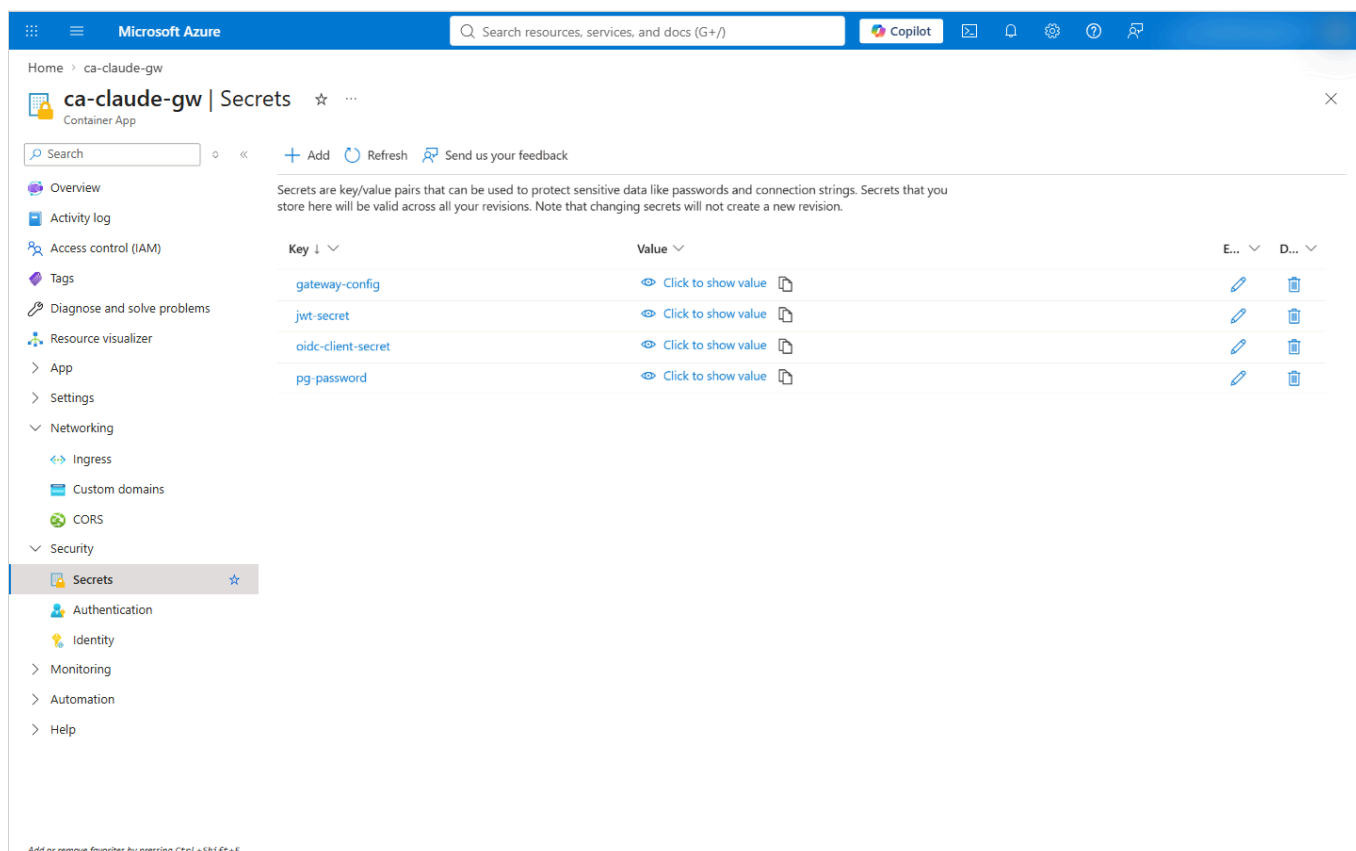
The readiness probe on /readyz fails when the store is unreachable ([health](#)). CLAUDE_GATEWAY_DRAIN_TIMEOUT_MS gives open streams two minutes when a replica stops, and the termination grace period is 10 seconds longer ([upgrades](#)). BUN_CONFIG_MAX_HTTP_REQUESTS is how many requests a replica sends upstream at once ([concurrent upstream requests](#)).

Azure portal

1. Render the configuration with the CLI tab's node scripts/admin/new-gateway-config.mjs --topology private.
2. **Container Apps** > **Create**: name ca-claude-gw, environment cae-claude-gw; **Container**: the image from acrclaudegw<suffix> with managed identity id-claude-gw, CPU and memory from the table; **Ingress**: enabled, **Limited to VNet**, target port 8080 ([portal quickstart](#)).
3. **Security** > **Secrets** > + **Add**, once for each secret in the table ([manage secrets](#)).
4. **Containers** > **Edit and deploy**: the command, arguments, environment variables, the secret volume at /etc/claude ([secret volumes](#)) and the probes ([health probes](#)); **Scale**: minimum 2, maximum 3, an HTTP rule with 150 concurrent requests ([scale rules](#)) > **Create**.



Screenshot of the Ingress page of ca-claude-gw: ingress enabled, traffic limited to the virtual network, HTTP, target port 8080, and the endpoint on the environment's default domain.



Screenshot of the Secrets page of ca-claude-gw listing gateway-config, jwt-secret, oidc-client-secret and pg-password with hidden values.

Azure CLI

Render the configuration, create the client secret, valid for 7 days like the script's default, and the session-signing secret, then write the app definition. The organization's credential policy sets the client secret's lifetime ([add a client secret](#)):

```

node scripts/admin/new-gateway-config.mjs --topology private
$end = [DateTime]::UtcNow.AddDays(7).ToString('yyyy-MM-ddTHH:mm:ssZ')
$oidcSecret = az ad app credential reset --id $appId --append --display-name gateway --end-date $end --query passw
ord -o tsv
$jwtSecret = [Convert]::ToBase64String([Security.Cryptography.RandomNumberGenerator]::GetBytes(48))
$environmentId = az containerapp env show -g $rg -n $environment --query id -o tsv
$config = (Get-Content config/gateway.azure-private.yaml | ForEach-Object { "          $_" }) -join "`n"

$definition = @"
location: $location
identity:
  type: UserAssigned
  userAssignedIdentities:
    "$identityId": {}
properties:
  environmentId: "$environmentId"
  workloadProfileName: Consumption
  configuration:
    activeRevisionsMode: Single
    ingress: { external: true, targetPort: 8080, transport: http, allowInsecure: false }
    registries:
      - { server: "$acr.azurecr.io", identity: "$identityId" }
    secrets:
      - name: gateway-config
        value: |
$config
      - { name: oidc-client-secret, value: "$oidcSecret" }
      - { name: jwt-secret, value: "$jwtSecret" }
      - { name: pg-password, value: "$pgPassword" }
  template:
    terminationGracePeriodSeconds: 130
    containers:
      - name: gateway
        image: "$image"
        command: [/usr/local/bin/claude]
        args: [gateway, --config, /etc/claude/gateway.yaml]
        resources: { cpu: 1.0, memory: 2Gi }
        env:
          - { name: CLAUDE_GATEWAY_LOG_LEVEL, value: info }
          - { name: AZURE_CLIENT_ID, value: "$identityClientId" }
          - { name: GATEWAY_PUBLIC_URL, value: "https://$app.$domain" }
          - { name: OIDC_ISSUER, value: "https://login.microsoftonline.com/$tenantId/v2.0" }
          - { name: OIDC_CLIENT_ID, value: "$appId" }
          - { name: ALLOWED_EMAIL_DOMAIN, value: "$allowedDomain" }
          - { name: GATEWAY_PG_USER, value: gatewayadmin }
          - { name: GATEWAY_PG_HOST, value: "$postgres.postgres.database.azure.com" }
          - { name: FOUNDRY_RESOURCE, value: "$foundry" }
          - { name: BUN_CONFIG_MAX_HTTP_REQUESTS, value: "256" }
          - { name: CLAUDE_GATEWAY_DRAIN_TIMEOUT_MS, value: "120000" }
          - { name: OIDC_CLIENT_SECRET, secretRef: oidc-client-secret }
          - { name: GATEWAY_JWT_SECRET, secretRef: jwt-secret }
          - { name: GATEWAY_PG_PASSWORD, secretRef: pg-password }
        probes:
          - { type: Startup, httpGet: { path: /healthz, port: 8080 }, initialDelaySeconds: 5, periodSeconds: 10, t
imeoutSeconds: 3, failureThreshold: 30 }
          - { type: Liveness, httpGet: { path: /healthz, port: 8080 }, periodSeconds: 15, timeoutSeconds: 3, failu
reThreshold: 3 }
          - { type: Readiness, httpGet: { path: /readyz, port: 8080 }, periodSeconds: 10, timeoutSeconds: 3, failu
reThreshold: 3 }
        volumeMounts:
          - { volumeName: gateway-config, mountPath: /etc/claude }
        volumes:
          - { name: gateway-config, storageType: Secret, secrets: [{ secretRef: gateway-config, path: gateway.yaml }]
}
scale:

```

```

    minReplicas: 2
    maxReplicas: 3
    rules:
      - { name: http-concurrency, http: { metadata: { concurrentRequests: "150" } } }
"@
[IO.File]::WriteAllText("$work\containerapp.yaml", $definition)

az containerapp create -g $rg -n $app --yaml "$work\containerapp.yaml" --output none
az containerapp show -g $rg -n $app --query "{latest:properties.latestRevisionName, ready:properties.latestReadyRevisionName, fqdn:properties.configuration.ingress.fqdn}"

```

The variables \$identityId, \$identityClientId, \$image, \$domain, \$pgPassword and \$appId come from steps 4 to 8, in the same PowerShell session; \$work\pg-password.txt holds the PostgreSQL password for a new session. The gateway admits a user who holds an app role and whose email claim ends with \$allowedDomain (config/gateway.azure-private.yaml:20-23). The app is ready when latest and ready name the same revision. The definition's schema is the Container Apps ARM specification ([ARM and YAML specification](#)), and the script builds the same definition (infra/azure-private/lib/AppDefinition.psm1:6-85).

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step app -MinReplicas 2 -MaxReplicas 3 -ConcurrentRequests 150
```

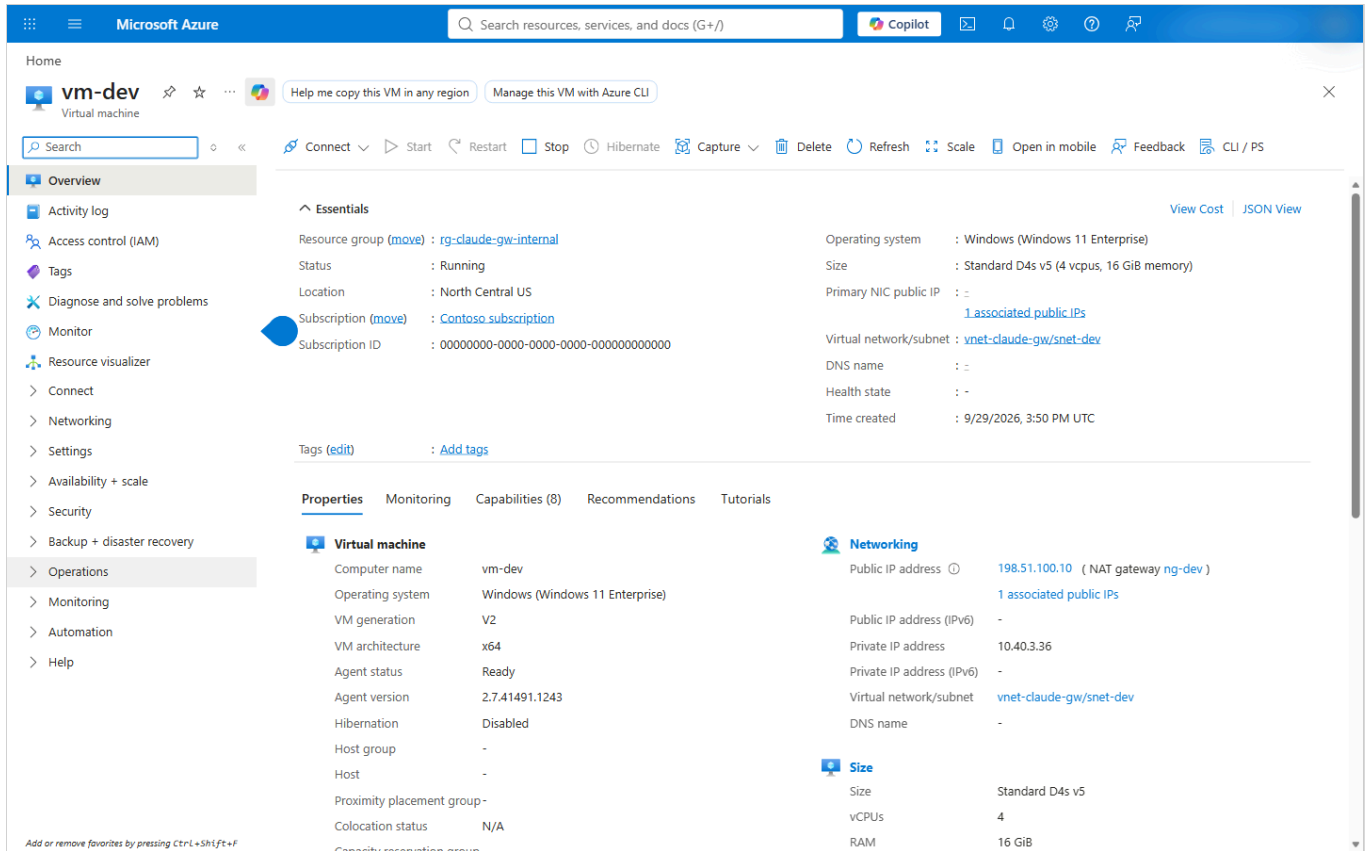
The step checks that config/gateway.azure-private.yaml is the rendering of its admin file, creates the secrets, writes the definition to a temporary folder, deploys it and waits for the revision to be ready (infra/azure-private/lib/Steps.App.psm1:81-127).

Step 10: Create a test machine in the network (optional)

A Windows 11 VM in snet-dev stands in for a developer machine on the corporate network. It has no public IP and is reached through Azure Bastion Developer, which needs no public IP or AzureBastionSubnet and connects to one VM at a time ([Bastion SKUs](#)). Run Command installs Claude Code, checked against the release checksum, and VS Code, and writes the machine policies that point Claude Code and Claude Desktop at the gateway ([Run Command](#); infra/azure-private/lib/Steps.Dev.psm1:9-33).

Azure portal

1. **Virtual machines** > **Create**: name vm-dev, region **North Central US**, image **Windows 11 Enterprise**, size Standard_D4s_v5; **Networking**: vnet-claude-gw / snet-dev, public IP **None**, NIC network security group **None**.
2. The VM > **Connect** > **Bastion**: with the Developer SKU, Bastion deploys on first connect ([Bastion Developer](#)).
3. The VM > **Operations** > **Run command** > **RunPowerShellScript**: the setup script of the CLI tab.



Screenshot of the overview of vm-dev: Windows 11 Enterprise in vnet-claude-gw / snet-dev with private IP address 10.40.3.36, no public IP address on its network interface, and outbound access through NAT gateway ng-dev.

The NAT gateway's public address in this screenshot is replaced with 198.51.100.10, a documentation address ([RFC 5737](https://www.rfc-editor.org/rfc/rfc5737)).

Azure CLI

The machine policies point Claude Code and Claude Desktop at this deployment's gateway; the file below holds them, with the host name from step 7. [Connect Claude Code, VS Code and Claude Desktop](#) describes both values:

```
$gatewayUrl = "https://$app.$domain"
$policy = @{ forceLoginMethod = 'gateway'; forceLoginGatewayUrl = $gatewayUrl; parentSettingsBehavior = 'merge' }
| ConvertTo-Json -Compress
[IO.File]::WriteAllText("$work\vm-setup.txt", @"
New-Item -Path 'HKLM:\SOFTWARE\Policies\ClaudeCode' -Force | Out-Null
New-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\ClaudeCode' -Name Settings -PropertyType String -Force -Value '$policy' | Out-Null
New-Item -Path 'HKLM:\SOFTWARE\Policies\Claude' -Force | Out-Null
New-ItemProperty -Path 'HKLM:\SOFTWARE\Policies\Claude' -Name bootstrapUrl -PropertyType String -Force -Value '$gatewayUrl/user/bootstrap' | Out-Null
"@)

# The VM's administrator password, for signing in through Bastion; keep it in the organization's credential store.
$vmPassword = ([Convert]::ToBase64String([Security.Cryptography.RandomNumberGenerator]::GetBytes(18)) -replace '[+/=]', '') + 'Aa1!'
[IO.File]::WriteAllText("$work\vm-password.txt", $vmPassword)
```

Create the VM and Bastion, then run the setup on the VM:

```
az vm create -g $rg -n vm-dev --location $location --image MicrosoftWindowsDesktop:windows-11:win11-25h2-ent:latest --size Standard_D4s_v5 --license-type Windows_Client --security-type TrustedLaunch --vnet-name $vnet --subnet snet-dev --public-ip-address '' --nsg '' --admin-username devadmin --admin-password "@$work\vm-password.txt" --output none
az network bastion create -g $rg -n bas-claude-gw --location $location --sku Developer --vnet-name $vnet
az vm run-command invoke -g $rg -n vm-dev --command-id RunPowerShellScript --scripts "@$work\vm-setup.txt"
```

In PowerShell, `''''` passes an empty value to the Azure CLI, and `"@file"` is quoted because `@` is special in PowerShell ([empty strings](#)). `Run Command` runs the file's lines as a PowerShell script on the VM ([Run Command](#)). The script's setup also installs Claude Code, checked against the release checksum, and VS Code (`infra/azure-private/lib/Steps.Dev.psm1:12-29`); on a machine set up by hand, Claude Code installs with Anthropic's installer ([set up Claude Code](#)).

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step devvm
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -ShowDevVmPassword
```

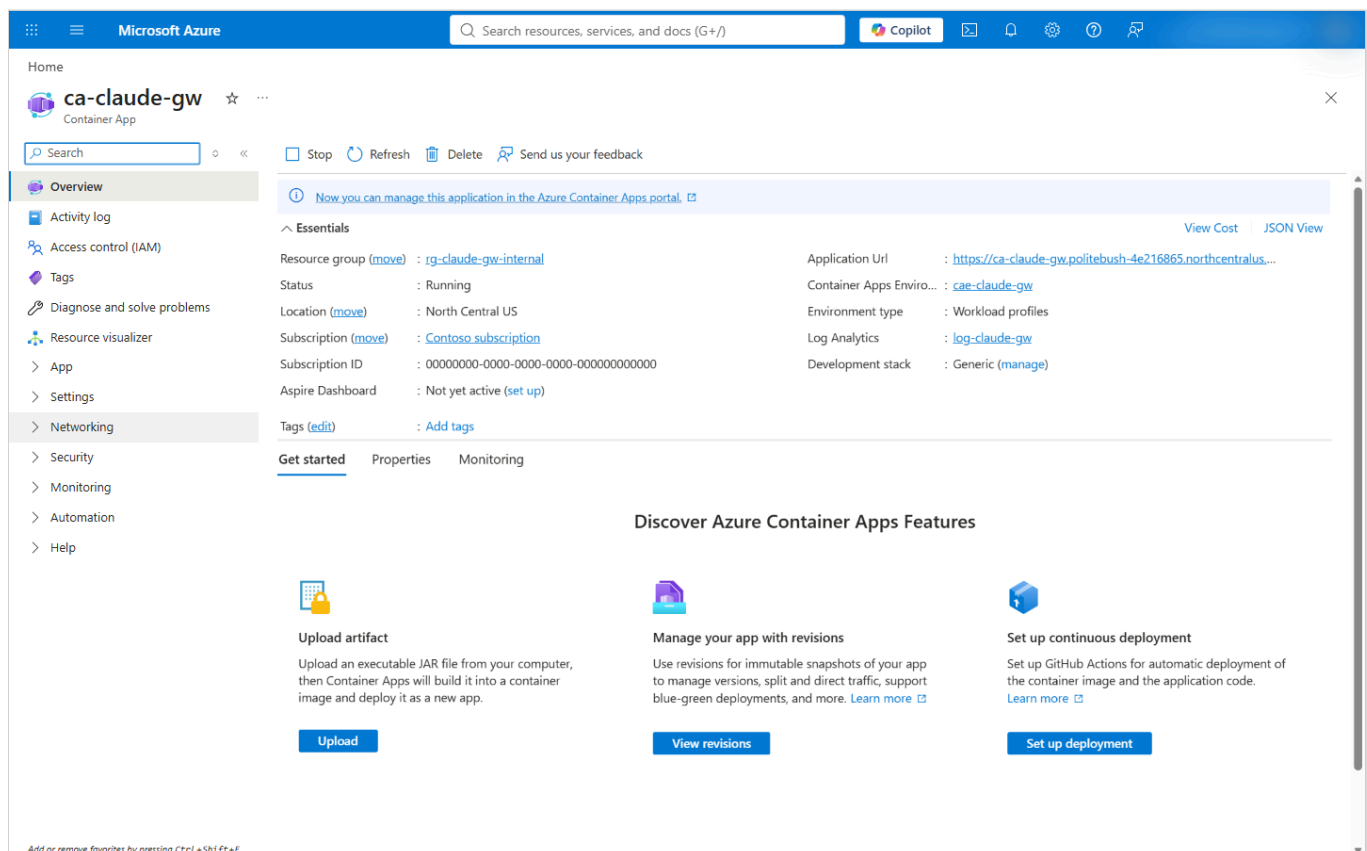
The script saves the VM password for the current Windows user with DPAPI; `-ShowDevVmPassword` prints the user name and the password (`infra/azure-private/lib/Steps.Dev.psm1:38-50`).

Step 11: Verify the deployment

From inside the network, every host name resolves to a private address and the gateway reports ready; from outside, Foundry refuses a request. The verify step runs these checks on the test machine with `Run Command` and on the operator's machine (`infra/azure-private/lib/Steps.Dev.psm1:80-121`).

Azure portal

1. The test machine > **Connect** > **Bastion**, then in PowerShell on the VM: `Resolve-DnsName ca-claude-gw.<default domain>` and `Invoke-WebRequest https://ca-claude-gw.<default domain>/readyz`.
2. **Container Apps** > `ca-claude-gw` > **Overview**: status **Running** and the application URL on the default domain.



The screenshot displays the Azure portal interface for the 'ca-claude-gw' Container App. The top navigation bar shows the Microsoft Azure logo and a search bar. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, App, Settings, Networking, Security, Monitoring, Automation, and Help. The main content area shows the 'Overview' page for the 'ca-claude-gw' Container App. It includes a search bar, a 'Stop' button, a 'Refresh' button, a 'Delete' button, and a 'Send us your feedback' link. The 'Essentials' section displays key information: Resource group (rg-claude-gw-internal), Status (Running), Location (North Central US), Subscription (Contoso subscription), Subscription ID (00000000-0000-0000-0000-000000000000), and Aspire Dashboard (Not yet active). The 'Tags' section shows 'Add tags'. The 'Get started' section includes 'Upload artifact', 'Manage your app with revisions', and 'Set up continuous deployment'. The 'Discover Azure Container Apps Features' section provides more details on these features.

Screenshot of the overview of `ca-claude-gw`: status **Running**, location North Central US, environment `cae-claude-gw`, application URL on the environment's default domain.

Azure CLI

The checks run on the test machine with `Run Command`, for this deployment's host name; the last command reads PostgreSQL's public access setting:

```
[IO.File]::WriteAllText("$work\vm-check.txt", @"
Resolve-DnsName $app.$domain -Type A
(Invoke-WebRequest -UseBasicParsing https://$app.$domain/readyz).StatusCode
"@)
az vm run-command invoke -g $rg -n vm-dev --command-id RunPowerShellScript --scripts "@$work\vm-check.txt"
az postgres flexible-server show -g $rg -n $postgres --query network.publicNetworkAccess -o tsv
```

Script

```
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step verify
```

```
PASS gateway host resolves to private addresses only (T-16)
PASS gateway /readyz answers 200 inside the VNet (T-16)
PASS PostgreSQL resolves to a private address (T-69)
PASS Foundry resolves to the private endpoint (T-68)
PASS Foundry refuses a request from outside the VNet: 403 (T-68)
PASS PostgreSQL public network access is Disabled (T-69)
addresses: gateway 10.40.1.231 postgres 10.40.3.4 foundry 10.40.2.6
```

Connect developers

Claude Code, the VS Code extension and Claude Desktop read the gateway's address from machine policy, which Intune or Group Policy delivers; a developer cannot configure the gateway sign-in in their own settings ([set the gateway URL](#)). The client setup, the sign-in, and the network each developer machine needs are in [Connect Claude Code, VS Code and Claude Desktop](#).

At the first sign-in, Claude Code shows the first 16 hexadecimal characters of the SHA-256 fingerprint of the gateway's certificate, and pins the certificate for the host name ([connect developers](#)). The fingerprint the verify step printed for this deployment starts with 2fbc49857fa6e209.

Run the whole deployment with the script

The script runs every step in order, creating what is missing (infra/azure-private/Deploy-Gateway.ps1:111-131):

```
# Print the commands without changing anything.
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Plan -ClaudeOrganizationName 'Contoso Ltd'
# Deploy.
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -ClaudeOrganizationName 'Contoso Ltd'
# Deploy chosen steps; later steps read the names earlier steps created.
pwsh -File infra/azure-private/Deploy-Gateway.ps1 -Step app,verify
```

Every parameter is in the script reference (docs/learn/reference-scripts.md). A re-run reuses existing resources and reports each with FOUND. It can also reconcile settings that drifted: Foundry's public access and key authentication, the DNS records, the role assignments, the Entra app's redirect URI and role assignment. It recovers secrets it cannot read, such as setting a new PostgreSQL password, runs the test machine's setup again, and updates the Container App with PATCH semantics (infra/azure-private/lib/Steps.Base.psm1:52-53; infra/azure-private/lib/Steps.App.psm1:44-48; infra/azure-private/lib/Steps.App.psm1:98-103; infra/azure-private/lib/Steps.Dev.psm1:73-74). -Plan lists what a re-run would change.

Results of the test deployment

Measured on 2026-09-29 with pwsh -File infra/azure-private/Deploy-Gateway.ps1, gateway image Claude Code 2.1.284. Verified: network isolation, name resolution, readiness, certificate trust, and Claude in Foundry answering the gateway's managed identity through the private endpoint from inside the gateway container (docs/UNKNOWNNS.md:68). From the test machine, Claude Code's /login showed the gateway's certificate fingerprint and got its code accepted by the gateway, which handed the browser to Microsoft Entra ID (docs/TEST-PLAN.md:280-281). Not yet verified: the Entra sign-in and a signed-in

request through the gateway, which waited on the tester's multifactor authentication, and the VS Code extension and Claude Desktop sign-ins.

Check	Result
Gateway host name inside the network	Resolves to 10.40.1.231 only; /readyz answers 200
PostgreSQL host name inside the network	Resolves to 10.40.3.4; public network access Disabled
Foundry host names inside the network	Resolve to 10.40.2.4, 10.40.2.5 and 10.40.2.6 (the private endpoint)
Foundry from outside the network	403 with a valid Entra ID token
Foundry from the gateway container, with the gateway's managed identity	claude-sonnet-5 answered 200 through the private endpoint, 10.40.2.6
Gateway certificate	Issuer Microsoft TLS G2 RSA CA OCSP 02, valid 2026-09-28 to 2027-01-06 (100 days), chain trusted by Windows (docs/UNKNOWNNS.md:67)
Region	North Central US; Central US refused a new environment with AKSCapacityHeavyUsage (docs/UNKNOWNNS.md:74)

The environment's default domain carries a certificate that the platform issues and renews. Each renewal changes the pinned certificate, and Claude Code shows each developer the trust prompt again ([connect developers](#)), so a production deployment uses a custom domain with a certificate whose rotation the organization schedules (docs/adr/0005-network-restricted-deployment.md:68).

Troubleshooting

The gateway's own boot and sign-in errors are in the [gateway troubleshooting table](#). These are specific to this deployment on Azure:

Symptom	Cause	Fix	Reference
az containerapp env create fails with AKSCapacityHeavyUsage: "Creating a new cluster is unavailable at this time in region ..."	The region has no capacity for a new environment	Create the network and the environment in another region; the Foundry account can stay where the quota is	docs/UNKNOWNNS.md:74
az postgres flexible-server create is refused for the region	The subscription is restricted from PostgreSQL flexible server in that region	Choose a region where az postgres flexible-server list-skus -l <region> lists the tier	docs/UNKNOWNNS.md:74
A Claude deployment is refused	The deployment has no modelProviderData block	Include organizationName, countryCode and industry	https://learn.microsoft.com/en-us/azure/developer/ai/how-to/deploy-claude-foundry#terms-of-use
az acr build output does not parse as JSON	Without --no-logs, the CLI prints the build log	Add --no-logs, then read status from the JSON	tests/azure-private.test.mjs:120-130
az network private-endpoint dns-zone-group show returns {} with exit code 0	The group does not exist; the CLI answers with an empty object	Treat an empty object as missing	infra/azure-private/lib/Az.psm1:85-87
/login: "Gateway hosts must be on your organization's private network; resolves to the public (or unrecognized) address "	The developer's DNS does not resolve the gateway's zone to the private IP	Forward the default domain's zone, or the custom domain's zone, to the private DNS zone through a DNS Private Resolver	https://code.claude.com/docs/en/claude-apps-gateway-on-aws#troubleshooting
/login gets 403; the gateway logs access.denied with reason	listen.trusted_proxies does not name the addresses the Container Apps ingress connects from, so the gateway checks the	Name the four ranges a workload profile environment reserves in trusted_proxies, as	docs/adr/0005-network-restricted-deployment.md:79-87

Symptom	Cause	Fix	Reference
ip_not_allowlisted and a client_ip in 100.100.0.0/16	ingress's address against access_control.allow_cidrs; /readyz still answers, because health probes are exempt	config/gateway.azure-private.yaml:14 does	
The revision does not become ready	/readyz fails while PostgreSQL is unreachable: a missing zone link, or a wrong password	Read the console log, then check the PostgreSQL zone's link and the pg-password secret	https://code.claude.com/docs/en/claude-apps-gateway-deploy#health
During a rollout the new replicas log remaining connection slots are reserved for roles with the SUPERUSER attribute	The old revision's replicas still hold their connections while the new ones open theirs, and together they exceed the server's user connections	Keep 2 × maximum replicas × store.max_connections within the server's user connections, or choose a larger PostgreSQL size	tests/azure-private.test.mjs:135-138
/login times out; the gateway logs Postgres is not answering, then, after store.readiness_grace_seconds, that /readyz reports not ready	The PostgreSQL server is stopped or unreachable. On 2026-09-29 an automated job of the test subscription stopped it (activity log: "Stops an existing server"); new sign-ins need the store, so the ingress stops routing to replicas that are not ready	Start the server with az postgres flexible-server start, and exclude it from automation that stops idle servers; the gateway reports ready again once PostgreSQL answers	https://code.claude.com/docs/en/claude-apps-gateway-deploy#outage-behavior
Inference fails with 401 or 403 from Foundry	The gateway's identity lacks Cognitive Services User on the account, or the assignment has not propagated	Assign the role on the Foundry account and retry after a few minutes	https://code.claude.com/docs/en/claude-apps-gateway-config#microsoft-foundry

Logs and telemetry

The gateway writes single-line JSON audit events and operational lines to stderr ([logs](#)). Container Apps sends them to the ContainerAppConsoleLogs_CL table of the environment's Log Analytics workspace ([log monitoring](#)):

```
az containerapp logs show -g $rg -n $app --type console --tail 50
```

```
ContainerAppConsoleLogs_CL
| where ContainerAppName_s == 'ca-claude-gw'
| where Log_s has '"evt":"inference"'
| project TimeGenerated, Log_s
```

An inference event names the user, the model, the upstream and the response status (tests/live/lib.mjs:177-185). Client metrics, logs and traces reach a collector through the gateway when telemetry.forward_to names one ([telemetry](#)).

Clean up resources

Deleting the resource group deletes every resource of this tutorial. The app registration is in Entra ID, outside the resource group, and a deleted Foundry account stays recoverable until it is purged ([purge a deleted resource](#)):

```
az group delete -n $rg --yes
az ad app delete --id $appId
az cognitiveservices account purge -g $rg -n $foundry -l $foundryLocation
```

The folder \$work holds the secret files of the CLI steps, including after a failed or interrupted step. A deployment that keeps running needs none of them, since the Container App holds its secrets, so the folder can go as soon as the gateway is ready and the VM password is in the organization's credential store:

```
Remove-Item -Recurse -Force $work
```

Next steps

[Connect Claude Code, VS Code and Claude Desktop](#)

- [Plan capacity for 25,000 developers](#)
- Configure models, roles and developer access (docs/learn/how-to-admin-configure.md)
- Script reference (docs/learn/reference-scripts.md)

Connect Claude Code, VS Code and Claude Desktop to the gateway

Deploy Windows client policies with Intune or Group Policy and connect Claude Code, its VS Code extension and Claude Desktop through the gateway's interactive sign-in.

This article configures Windows machines for the network-restricted Azure deployment, with Microsoft Entra ID sign-in and Microsoft Foundry upstreams. Production uses a custom domain with the organization's certificate (docs/adr/0005-network-restricted-deployment.md:65-74); the example URL is `https://claude-gateway.corp.contoso.com`. A custom domain replaces the default host name in four places: a private DNS record, a certificate bound to the Container App ([custom domains and certificates](#)), `GATEWAY_PUBLIC_URL`, which is the gateway's `listen.public_url`, and the Entra redirect URI `<origin>/oauth/callback`. `Deploy-Gateway.ps1` sets the last two from the default host name (infra/azure-private/lib/AppDefinition.psm1:16; infra/azure-private/lib/Steps.App.psm1:44). Until that change, the client values use `https://ca-claude-gw.<environment default domain>`.

Client	Machine setting	Sign-in	Minimum version	Source
Claude Code CLI	<code>forceLoginMethod</code> , <code>forceLoginGatewayUrl</code> , <code>parentSettingsBehavior</code> in Claude Code managed settings	/login, then gateway device-code approval in a browser	Claude Code 2.1.195	Prerequisites , login keys
Claude Code extension for VS Code	The same machine-level Claude Code settings	Panel sign-in and browser authorization; gateway-specific screens are not documented	VS Code 1.94.0; no extension-specific gateway minimum is documented	Extension prerequisites , first use , policy surfaces
Claude Desktop	<code>bootstrapUrl</code> under <code>HKLM\SOFTWARE\Policies\Claude</code> ; keep the Claude Code merge opt-in too	Its own gateway device-code sign-in; not the CLI session	Desktop 1.10628.0 for <code>bootstrapUrl</code> ; gateway 2.1.203	Bootstrap keys , Desktop connection

Prerequisites

- Complete [Deploy the network-restricted gateway](#); use its HTTPS hostname and a trusted certificate ([TLS requirements](#)).
- Connect the machine and sign-in browser to the corporate network or VPN, with private-zone DNS forwarding ([DNS Private Resolver](#)).
- Install the listed clients; check the standalone CLI with `claude --version`. The extension bundles its own CLI ([VS Code prerequisites](#)).
- A `CLAUDE_CODE_USE_*` provider variable, such as `CLAUDE_CODE_USE_FOUNDRY`, selects that provider directly, without the gateway sign-in ([set the gateway URL](#)).
- Have Node.js 22 or later for the generator (docs/learn/reference-scripts.md:13), and administrator rights or device management for HKLM ([delivery mechanisms](#)).
- Assign a user, or groups for the 25,000-developer rollout, to `Gateway.Standard` or `Gateway.Premium` on the gateway's enterprise application (config/gateway.azure-private.yaml:22-23). This deployment names its app registration `claude-apps-gateway-private-<suffix>` (infra/azure-private/Deploy-Gateway.ps1:84). In the Microsoft Entra admin center, open **Enterprise applications**, select that gateway application, then **Users and groups** > **Add user/group**. Select the user or group, choose **Select a role**, select the gateway role, and select **Assign**. Group-based assignment requires **Microsoft Entra ID P1 or P2**; nested group memberships do not grant access through the assignment ([assign users, groups and roles](#)).

Network requirements for developer machines

The deployment constructs `https://ca-claude-gw.<environment default domain>` and links a private DNS zone with a wildcard A record pointing to the environment's static IP (infra/azure-private/lib/Steps.App.psm1:23-31). Forward that zone

from corporate DNS to an Azure DNS Private Resolver inbound endpoint; give a custom hostname private resolution too ([inbound endpoints](#)).

From a developer machine, check resolution and connectivity:

```
Resolve-DnsName claudе-gateway.corp.contoso.com
Test-NetConnection claudе-gateway.corp.contoso.com -Port 443
```

Every address must be RFC 1918, CGNAT 100.64.0.0/10, link-local, IPv6 ULA fc00::/7, or loopback; a public A or AAAA answer fails /login ([private-network requirement](#)).

Destination	Port	Used for	Source
Gateway hostname, resolving to its private IP	TCP 443	CLI and browser sign-in, Desktop bootstrap and model requests	Gateway prerequisites
login.microsoftonline.com and the documented authentication hosts, including *.msauth.net and *.aadcdn.msftauth.net	TCP 443	Entra browser sign-in and its page resources; use the full authentication list for your environment	Microsoft authentication endpoints
claude.ai	TCP 443	Downloading the native CLI installation script, if using Anthropic's installer	Install Claude Code
downloads.claude.ai	TCP 443	CLI native binaries and updates; Desktop session components, model catalog and update binaries	CLI network access , Desktop egress
claude.ai, api.anthropic.com; alternatively releases.claude.com with updateViaUpdatesHost	TCP 443	Desktop update feeds when auto-updates are enabled	Desktop egress

Export Desktop's **Egress** list for all enabled telemetry, services and connectors; this table is not a complete feature allowlist ([Desktop egress](#)). The offline installer bundles session components, but its catalog still uses downloads.claude.ai unless disabled or mirrored ([MDM egress](#)).

Add the gateway hostname to NO_PROXY, retaining other entries: HTTPS-proxied CLI requests skip certificate pinning ([pinning](#)). Desktop's app follows its own OS or managed proxy configuration, not the CLI shell variable ([Desktop proxy](#)).

Generate the policy files

From the repository root, use an empty output directory with the generator in Generate client payloads (docs/learn/how-to-admin-configure.md):

```
node scripts\admin\new-client-policy.mjs --gateway-url https://claude-gateway.corp.contoso.com --fast-model claude-sonnet-5 --out .\client-policy
```

Options and outputs: scripts/admin/new-client-policy.mjs:155-179. [Windows policy locations](#) define the destinations below.

Output	Purpose
login\managed-settings.json	File-based policy; install as C:\Program Files\ClaudeCode\managed-settings.json
login\ClaudeCode-HKLM.reg	The same JSON as the Settings string value under HKLM\SOFTWARE\Policies\ClaudeCode
login\ClaudeCode.mobileconfig	macOS configuration profile; not used on Windows
developer\	Alternative apiKeyHelper profile and instructions; do not deploy alongside forced gateway login

The login payload contains these three keys (scripts/admin/new-client-policy.mjs:46):

```
{
  "forceLoginMethod": "gateway",
  "forceLoginGatewayUrl": "https://claude-gateway.corp.contoso.com",
  "parentSettingsBehavior": "merge"
}
```

The login keys require an administrator-managed source on the machine, not user/project settings, HKCU or the gateway's `managed.policies[].cli` ([client-side settings](#)). `--fast-model` affects only the alternative developer folder's install command (`scripts/admin/new-client-policy.mjs:97-98`).

The generator does not write Desktop's policy. Add this separate `REG_SZ` value ([Desktop value types](#)):

```
reg add "HKLM\SOFTWARE\Policies\Claude" /v bootstrapUrl /t REG_SZ /d "https://claude-gateway.corp.contoso.com/user/bootstrap" /f
```

Equivalent `.reg` content:

```
Windows Registry Editor Version 5.00

[HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Claude]
"bootstrapUrl"="https://claude-gateway.corp.contoso.com/user/bootstrap"
```

The VM setup downloads `claude.exe`, checks its release SHA-256, writes both HKLM policies and installs VS Code (`infra/azure-private/lib/Steps.Dev.psm1:17-29`).

Deliver the policy with Intune or Group Policy

Registry policy replaces the managed settings file by default. Keep all three keys and other required controls in one complete JSON value ([source precedence](#)).

Intune

1. Use a PowerShell platform script to write both HKLM values, or a system-context Win32 app to import the `.reg` and set Desktop's value ([Intune Management Extension](#), [Win32 install context](#)). The packaged installer can run `reg import .\client-policy\login\ClaudeCode-HKLM.reg`, followed by the Desktop command above.
2. For a platform script, open **Devices > Scripts and remediations > Platform scripts > Add > Windows 10 and later**. Set **Run this script using the logged on credentials** to **No** (system context) and **Run script in 64-bit PowerShell host** to **Yes** ([script policy settings](#)).
3. Assign to a pilot device group; review **Device status** before expanding ([monitor run status](#)).

Group Policy

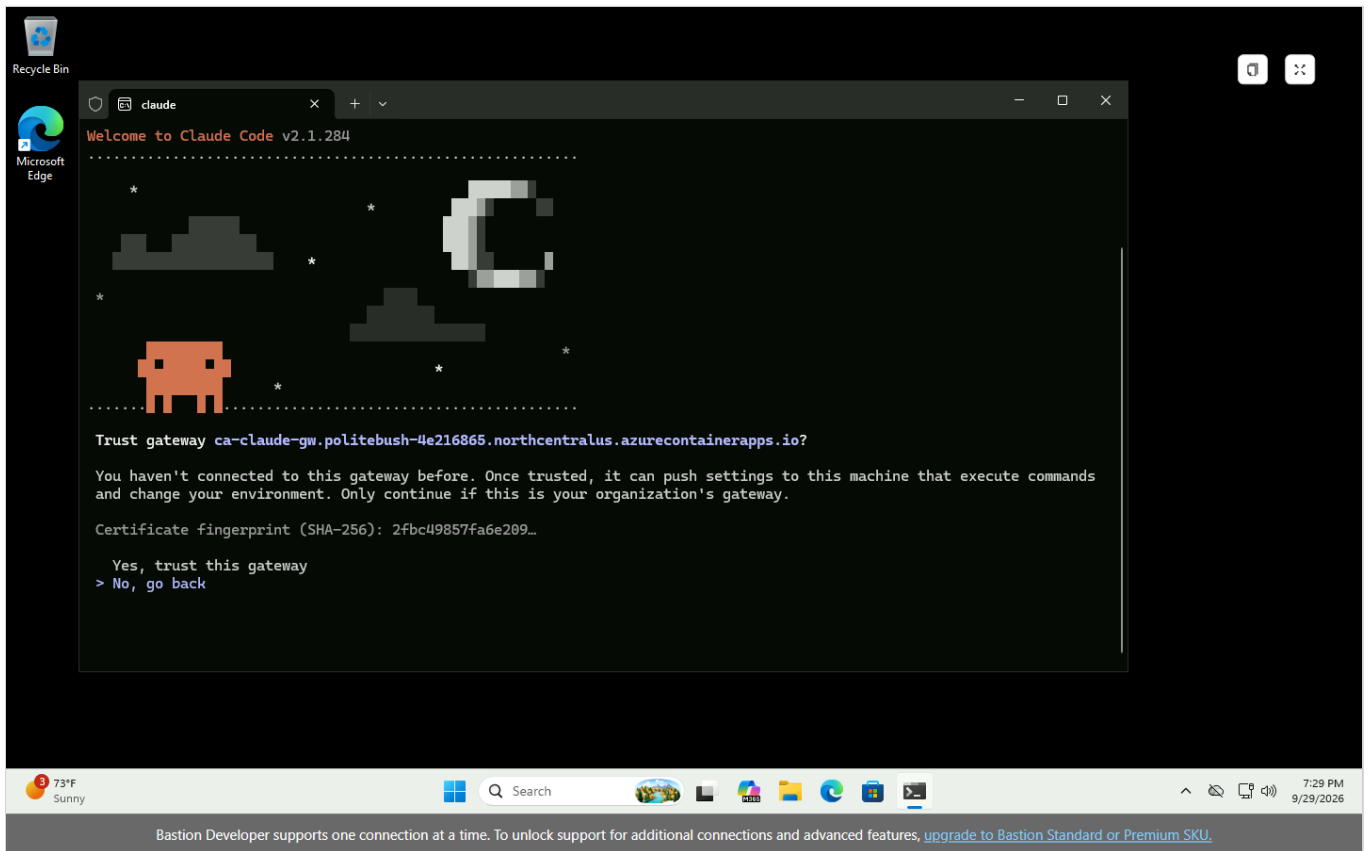
1. Edit the device GPO in Group Policy Management. Open **Computer Configuration > Preferences > Windows Settings > Registry**.
2. Add **Registry Item** entries: action **Update**, hive **HKEY_LOCAL_MACHINE**, type **REG_SZ**. Set `SOFTWARE\Policies\ClaudeCode / Settings` to the complete JSON; set `SOFTWARE\Policies\Claude / bootstrapUrl` to `https://claude-gateway.corp.contoso.com/user/bootstrap` ([configure a Registry item](#)).

For 25,000 developers, stage assignments with sign-in instructions. NAT/VPN users share per-address sign-in limits ([large rollouts](#)). The sample sets 1,000 starts and 100 submissions per address per 600 seconds, not fleet-wide capacity (`config/gateway.azure-private.yaml:71-73`).

Sign in from Claude Code

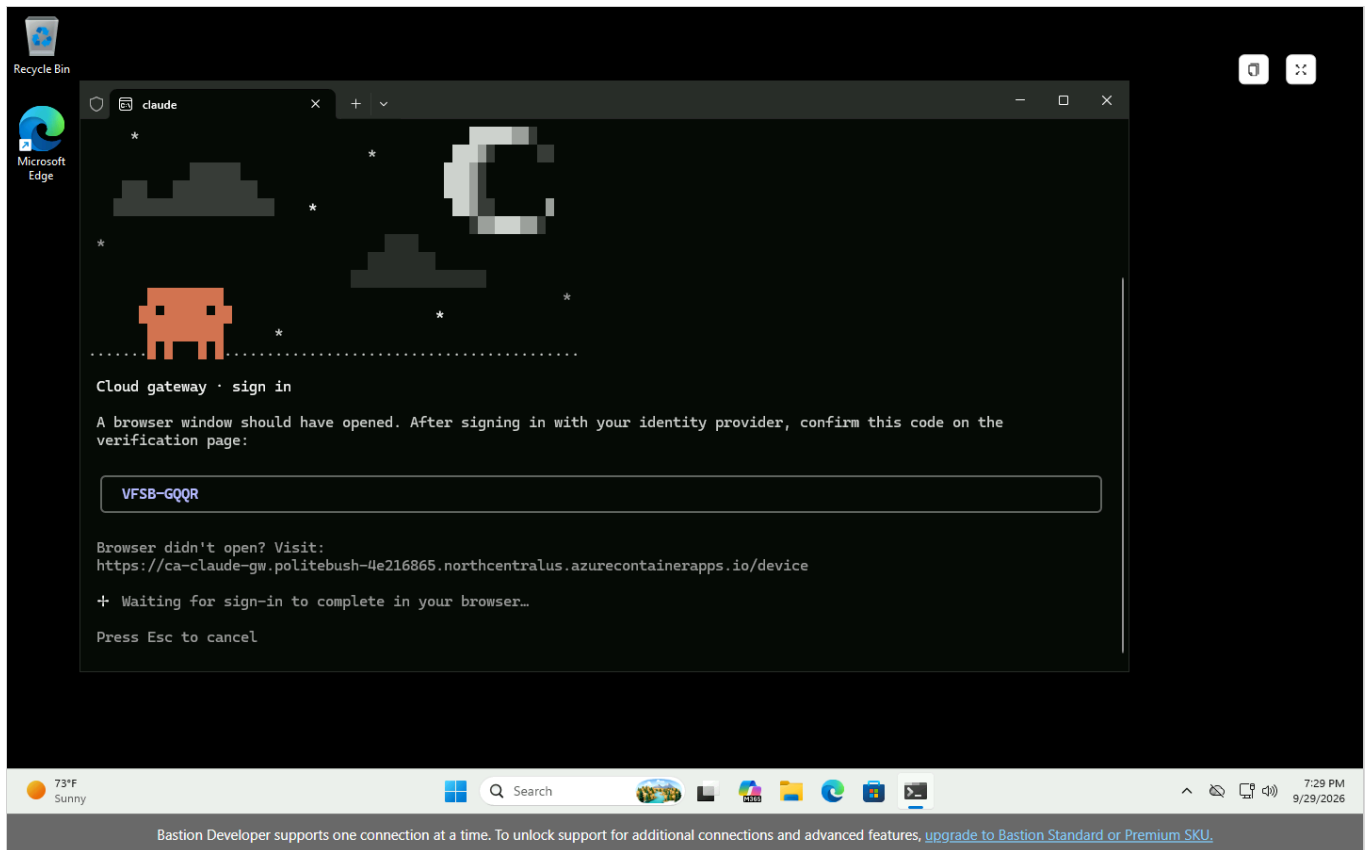
1. Open PowerShell and run `claude`.
2. Enter `/login`; the **Cloud gateway** screen has the URL filled in. Press **Enter** ([login screen](#)).
3. Compare the first-connect fingerprint with the administrator's published value: the first 16 lowercase hex characters of the leaf certificate's SHA-256, without colons. Accept only a match ([pinning](#)).
4. Open the verification link and confirm the code on the gateway's page. Sign in to Entra with the assigned work account; the browser returns to `/oauth/callback` ([device flow callback](#)).
5. Run `/status`; **Setting sources** shows gateway policy as remote managed settings. Use `/model` for allowed models ([policy check](#), [models](#)).

On the test machine of the [tutorial](#), Claude Code 2.1.284 showed these screens on 2026-09-29 (docs/TEST-PLAN.md:280-281). After **Enter** on the Cloud gateway screen, it asks to trust the gateway and shows the start of the certificate's SHA-256 fingerprint:



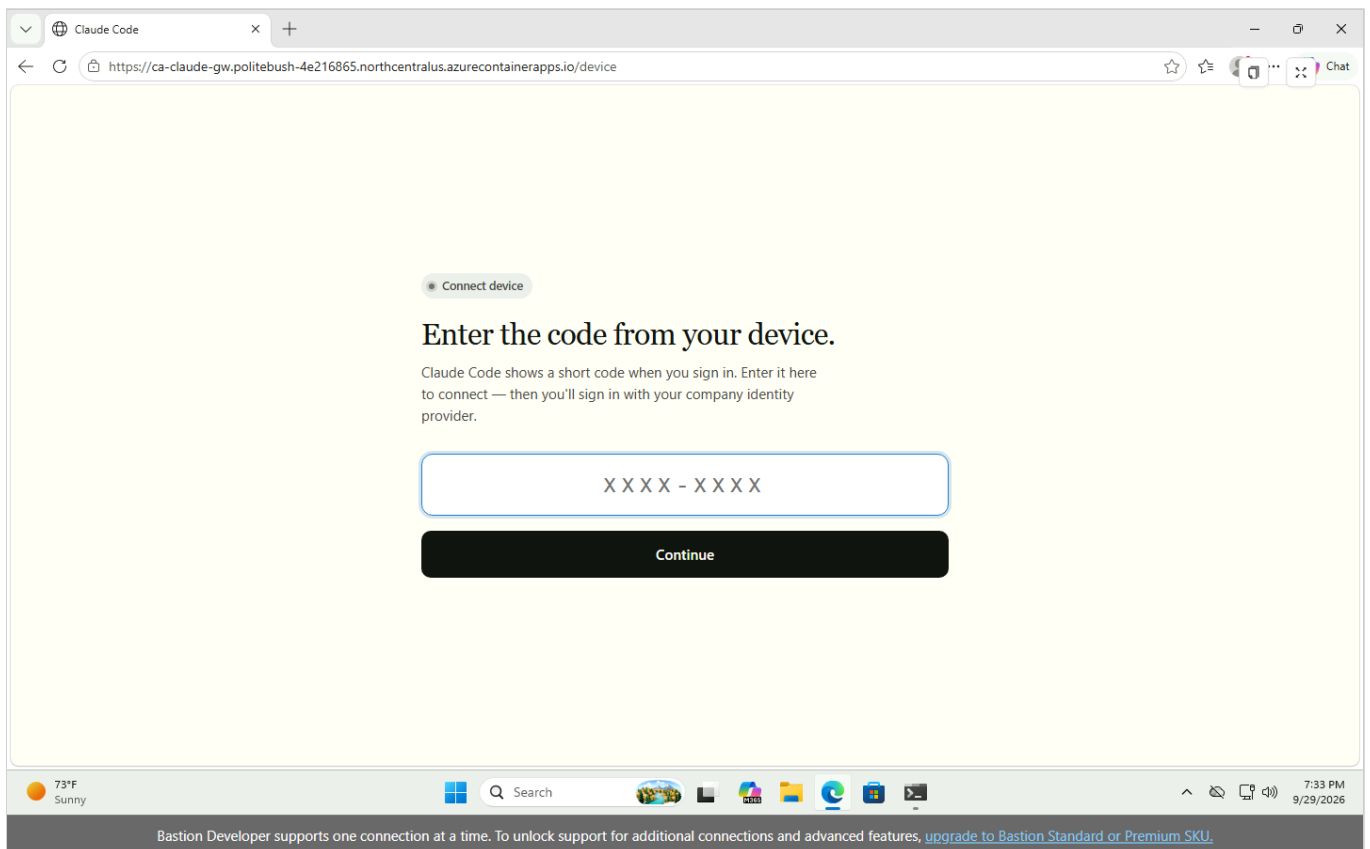
Screenshot of Claude Code asking Trust gateway ca-claude-gw on the environment's default domain, with the certificate fingerprint 2fbc49857fa6e209 and the choices Yes, trust this gateway and No, go back.

After the trust, it opens the browser, shows the code and the gateway's verification address, and waits:



Screenshot of Claude Code's Cloud gateway sign-in screen with a one-time code, the address of the gateway's device page, and Waiting for sign-in to complete in your browser.

The gateway's device page takes the code, then sends the browser to Microsoft Entra ID for the sign-in; a mistyped code is refused on the page:



Screenshot of the gateway's device page in Microsoft Edge on the test machine: Enter the code from your device, a code field and a Continue button.

Administrators can publish the fingerprint from the served leaf certificate file with OpenSSL in PowerShell:

```
$fingerprint = ((openssl x509 -noout -fingerprint -sha256 -in .\cert.pem) -split ' ', 2)[1].Replace(':', '').ToLowerInvariant()
$fingerprint
$fingerprint.Substring(0, 16)
```

This formats Anthropic's [OpenSSL command](#) for PowerShell. The verify step also prints sha256 (infra/azure-private/lib/Steps.Dev.psm1:94). The CLI pins the leaf per hostname; every rotation prompts every developer again. Publish the replacement fingerprint.

Over SSH, run `/login` remotely and open the link on a laptop that can reach the gateway ([remote machines](#)). Client 2.1.275 added optional email confirmation, but this gateway returns no email field, so no such confirmation appears ([connection behavior](#)).

Sign in from the VS Code extension

1. Deploy the same machine policy; the extension's bundled CLI reads managed sources ([runtime](#), [policy surfaces](#)).
2. Open **Extensions** with **Ctrl+Shift+X**, install **Claude Code**, then open its Spark-icon panel ([install](#), [open panel](#)).
3. On first use, select **Sign in** and complete browser authorization ([first use](#)).

The guide does not specify gateway-only screens, panel behavior for `forceLoginMethod`, or CLI credential reuse. `disableLoginPrompt` skips prompts for direct third-party provider setups; leave its default `false` here ([extension settings](#)). The integrated terminal's `claude` command needs the standalone CLI, separate from the panel ([extension prerequisites](#)).

Connect Claude Desktop

1. Both roles opt in with `desktop: {}` (config/gateway.azure-private.yaml:55-65); without it, `/user/bootstrap` returns 404 ([Desktop overlay](#)).
2. Deploy `bootstrapUrl` directly under `HKLM\SOFTWARE\Policies\Claude`. Since Desktop 1.19367.0, HKLM policy excludes HKCU policy entirely; keep configuration in one hive ([registry precedence](#)).
3. Quit and reopen Desktop; configuration is read at launch ([how keys are read](#)).
4. Complete Desktop's browser sign-in. With only `bootstrapUrl`, and no `bootstrapIdc` or header credentials, Desktop uses device-code mode against the bootstrap server's origin ([bootstrap authentication](#)).
5. Cowork and Code send model requests through the gateway; CLI and Desktop sign-ins are separate. Enable Chat with `chatTabEnabled: true` in Desktop policy or the desktop overlay (gateway 2.1.227+ for the overlay) ([Desktop tabs](#)).

Retain `parentSettingsBehavior: "merge"` in the selected Claude Code managed source on Desktop-only machines: it lets a gateway egress allowlist reach embedded sessions. This repository sets model lists, not an egress allowlist (config/gateway.azure-private.yaml:54-65; [Desktop policy delivery](#)). The MDM-only `trustBootstrapDelivery` (Desktop 1.26832.0+) skips consent for bootstrap-delivered sign-in targets, endpoints, helper scripts and connectors, not sign-in ([bootstrap controls](#)).

What the gateway enforces after sign-in

- **Models:** Standard gets Sonnet 5; Premium adds Opus 5 (config/gateway.azure-private.yaml:55-65). The picker follows `availableModels`; other model requests return 400 ([model enforcement](#)).
- **CLI policy:** locked settings cannot be overridden locally; policy loads at startup and refreshes hourly, with some changes deferred. A signed-in CLI exits at startup after about 10 seconds if the gateway is unreachable ([policy and startup](#)).
- **Credentials:** clear conflicting API credentials; forced gateway sign-in does not use a leftover API key or `claude.ai` login ([credential errors](#)).
- **Lifetime:** IdP deprovisioning ends the session within `ttl_hours` when refresh fails; here it is one hour (config/gateway.azure-private.yaml:27; [deprovisioning](#)).
- **Sign-out:** `/logout` deletes the local credential. Client 2.1.275 supports revocation, but this server advertises no endpoint: logout is local only ([sign-out](#)).
- **CI:** no unattended service-token flow exists ([CI limitations](#)).

Troubleshoot

Symptom	Cause	Fix	Reference
Gateway hosts must be on your organization's private network; <host> resolves to the public (or unrecognized) address <ip>	At least one DNS answer is public	Correct corporate DNS forwarding and A/AAAA records; use the private route	Private-address refusal
No gateway option in /login	Missing login keys, or keys in user settings/HKCU	Deploy all three keys in the selected machine-managed source, then start Claude Code again	Set the gateway URL
The gateway is limiting sign-in attempts right now	Per-address sign-in limit reached; this wording requires client 2.1.274+	Retry after the window; operators check trusted_proxies before sizing sign-in limits for shared addresses	Large rollouts
Desktop cannot fetch /user/bootstrap (404), or is not configured	Missing matching desktop policy, wrong bootstrapUrl, or unread policy	Check the matching policy's desktop: {}, the HKLM value and gateway URL; quit and reopen Desktop	Desktop overlay, registry delivery
Fingerprint prompt after a certificate change	Leaf certificate no longer matches the pin	Compare the new published fingerprint; for repeated unexpected changes, check ingress certificates and TLS interception	Certificate pinning
Administrator policy requires a Cloud gateway sign-in	API environment variables, an apiKeyHelper, or a saved Console API key conflict with policy	Clear those credentials; use claude auth logout for a saved key, then start claude and run /login	Error reference
Proxy refusal, or certificate pin checks skipped	An HTTPS proxy is on the CLI's gateway route	Add claude-gateway.corp.contoso.com to NO_PROXY and relaunch; retain other bypass entries	Proxy and pinning
Gateway rejects requests from a corporate subnet	The source is outside access_control.allow_cidrs	Have the operator configure the admitted developer networks; the sample admits 10.40.0.0/16	config/gateway.azure-private.yaml:68

Related content

- Overview (docs/learn/overview.md)
- Configure models, roles and developer access (docs/learn/how-to-admin-configure.md)
- Developer profile quickstart (docs/learn/quickstart-developer.md)
- Troubleshoot the developer profile (docs/learn/troubleshoot.md)
- [Deploy the network-restricted gateway](#)
- [Plan capacity for 25,000 developers](#)

Plan capacity for 25,000 developers

Size sign-in limits, gateway replicas, PostgreSQL, Foundry quota, identity and private connectivity for a rollout to 25,000 developers.

A rollout has six separate capacity boundaries: sign-in limits, gateway replicas, PostgreSQL connections, Foundry quota, identity, and network/DNS/certificate management. Developer count alone does not determine inference load. This article covers the internal Container Apps environment, private PostgreSQL store and private-endpoint Foundry accounts in docs/adr/0005-network-restricted-deployment.md:55-77. Worked examples are assumptions, not measured capacity.

Limits at a glance

Defaults and repository inputs were checked on September 29, 2026; they are not capacity benchmarks.

Limit	Default	Setting in this repository	Source
Sign-in starts, per client address	30 per 600 seconds	rate_limits.device_authorization: 1,000 per 600 seconds	HTTP tuning ; config/gateway.azure-private.yaml:72
Code submissions, per client address	10 per 600 seconds	rate_limits.device_verify: 100 per 600 seconds	HTTP tuning ; config/gateway.azure-private.yaml:73
Concurrent upstream requests, per replica	256	-MaxUpstreamRequests sets BUN_CONFIG_MAX_HTTP_REQUESTS	Upstream concurrency ; infra/azure-private/lib/AppDefinition.psm1:25
HTTP scale-out target	10 concurrent requests	-ConcurrentRequests: 150	HTTP scaling ; infra/azure-private/Deploy-Gateway.ps1:42
PostgreSQL pool, per replica	5 connections	store.max_connections: 5	Store ; config/gateway.azure-private.yaml:33
Readiness grace after a store outage	0 seconds	store.readiness_grace_seconds: 300	Store ; config/gateway.azure-private.yaml:35
Gateway session lifetime	1 hour	session.ttl_hours: 1	Session ; config/gateway.azure-private.yaml:27
Gateway shutdown drain	25 seconds	120 seconds; container termination grace 130 seconds	Upgrades ; infra/azure-private/lib/AppDefinition.psm1:28; infra/azure-private/lib/AppDefinition.psm1:59

Sign-in rate limits

These limits apply to sign-in, not inference. Without `listen.trusted_proxies`, the gateway counts the ingress address rather than developer addresses ([large rollouts](#)). For evenly distributed sign-ins, attempts per address per window = developers / addresses × window / rollout duration. Set `max` = round up(estimated attempts × 2): the factor of 2 allows retries and multiple clients. Calculate starts and browser code submissions separately if their network paths differ.

- **Assumption: distinct private addresses.** Each of the 25,000 developers keeps an individual private address. Their CLI, extension, Desktop and browser attempts stay below 30 starts and 10 submissions per 600 seconds. Defaults suffice; total headcount does not combine their buckets.
- **Assumption: shared addresses.** Developers share 10 NAT/VPN egress addresses and sign in evenly over 2 hours: $25,000 / 10 = 2,500$ developers per address; $120 \text{ minutes} / 10 \text{ minutes} = 12$ windows; about 208.3 sign-ins per window. Doubling gives about 416.7; rounding up to **500** for each limit provides an example rollout setting.

Measure the busiest `client_ip/window` in [sign-in audit events](#). The repository's 100 code submissions would throttle this example even though its 1,000 starts suffice.

Set `deployment.signInsPerAddress` and `deployment.codeSubmissionsPerAddress` to 500 for this example (config/gateway-admin.azure-private.json:38-39). Windows remain 600 seconds (config/gateway.azure-private.yaml:72-73). Render, then

check, from the repository root:

```
node scripts\admin\new-gateway-config.mjs --topology private
node scripts\admin\new-gateway-config.mjs --topology private --check
```

Options are defined at scripts/admin/new-gateway-config.mjs:239-245; rendering does not deploy the change. device_verify protects against guessing codes: raise it only as needed. Lower limits after rollout if refresh tokens enable silent renewal; otherwise size for repeated sign-ins ([large rollouts](#)).

Gateway replicas

Use [Little's law](#): average concurrency = requests/second × average seconds open. Estimate the busiest sustained interval, then test bursts. Count model requests, including tool turns, not user prompts.

Input	Assumption for this example	How to measure
Developers active at peak	20% of 25,000 = 5,000	Distinct sub values in inference events during the peak interval
Requests per active developer	2 per minute	Model request count divided by active developers and interval duration
Average time a request stays open	30 seconds	Time from request start to stream end in the client/load generator

Audit fields are recorded in docs/UNKNOWNNS.md:55; the target is 150, below 256 upstream slots (infra/azure-private/Deploy-Gateway.ps1:42-43).

```
request rate = 25,000 × 20% × 2 = 10,000 requests/minute
concurrency = (10,000 / 60) × 30 = 5,000 open requests
replicas = ceiling(5,000 / 150) = 34
```

This is a floor, not a production recommendation. Allow for bursts, uneven routing, overlap and zone loss. -MinReplicas, -MaxReplicas and -ConcurrentRequests control the range/target (infra/azure-private/lib/AppDefinition.psm1:76-80). The HTTP target is a scaling signal, not a hard cap ([HTTP scaling](#)).

A stream occupies an upstream slot until it ends; queued bodies also consume memory. -MaxUpstreamRequests sets BUN_CONFIG_MAX_HTTP_REQUESTS (1–65,535). Measure CPU/memory before raising it, and keep the HTTP target below it ([upstream concurrency](#)).

CPU/memory capacity remains unmeasured (U-58, docs/UNKNOWNNS.md:69). -Cpu/-Memory request 1 vCPU/2 GiB (infra/azure-private/Deploy-Gateway.ps1:44-45). Use [load test mode](#) on an isolated deployment with its own empty database, never a gateway developers use. It returns canned replies without Foundry; x-load-test-user simulates distinct users. The mode needs gateway 2.1.282+; CPU estimates before 2.1.283 are substantially lower. Even newer estimates omit upstream encryption.

Hold one replica fixed; vary duration, body size and concurrency; measure queue onset and CPU/memory growth per open request with [replica metrics](#). Include planned spend enforcement, then test autoscaling and real Foundry.

Create with -ZoneRedundant where supported; keep at least 2 minimum replicas, more if the peak cannot wait for scaling ([zone distribution](#)). Azure allows [1,000 replicas per revision](#). The Consumption-core [quota](#) sums all active replicas' cores across apps; allocations vary. Read it before selecting -MaxReplicas:

```
az containerapp env list-usages --resource-group "<resource-group>" --name "<environment>"
```

Request more through the environment's **Quota** page. The example needs 34 × 1 = 34 cores before overlap/other apps. The deployment's allocation remains unchecked (U-60, docs/UNKNOWNNS.md:71).

CLAUDE_GATEWAY_DRAIN_TIMEOUT_MS is 120,000 ms; termination grace is 130 seconds (infra/azure-private/lib/AppDefinition.psm1:28; infra/azure-private/lib/AppDefinition.psm1:59). Keep grace at least 5 seconds beyond drain; unfinished streams are cut ([upgrades](#)). Azure documents a [240-second HTTP ingress timeout](#). A 403.5-second stream

completed in the test deployment (U-40, docs/UNKNOWNNS.md:51), not a guarantee at every load. Retest long streams on the private deployment, including scale-in and upgrades.

PostgreSQL

Budget for the old and new revisions of a rollout, which hold connections at the same time: $2 \times \text{replicas} \times \text{store.max_connections} + \text{other connections} \leq \text{max_connections} - \text{reserved connections}$. On 2026-09-29 a rollout of 2 replicas at 10 connections each, next to the old revision's 2, logged remaining connection slots are reserved for roles with the SUPERUSER attribute on Burstable B1ms (tests/azure-private.test.mjs:135-138). Pools are [per replica, not per developer](#). Azure currently reserves 15 connections. Read `max_connections`, `reserved_connections` and `superuser_reserved_connections`; defaults do not automatically increase after a SKU change ([PostgreSQL limits](#)).

SKU	Default maximum connections	Maximum user connections
Burstable B1ms	50	35
Burstable B2s	429	414
General Purpose D2ds_v5	859	844
General Purpose D4ds_v5	1,718	1,703
General Purpose D8ds_v5	3,437	3,422
General Purpose D16ds_v5	5,000	4,985

The example's $2 \times 34 \times 10 = 680$ connections, a rollout of 34 replicas at 10 connections each, fit D2ds_v5's 844 before other clients, but do not establish CPU/I/O capacity. Set the pool through `deployment.storeConnectionsPerReplica` (config/gateway-admin.azure-private.json:40).

The B1ms test ceiling is `-MaxReplicas 3` with 5 connections per replica: $2 \times 3 \times 5 = 30$ of 35 maximum user connections (infra/azure-private/Deploy-Gateway.ps1:38-41). A test checks this default budget (tests/azure-private.test.mjs:132-142). Production guidance is General Purpose with zone-redundant HA (docs/adr/0005-network-restricted-deployment.md:69; [HA support](#)). `-PostgresSku`, `-PostgresTier` and `-PostgresZonalResiliency` apply to new servers, not upgrades (infra/azure-private/lib/Steps.Base.psm1:92-100).

Built-in [PgBouncer](#) uses port 6432 on General Purpose/Memory Optimized tiers. Compatibility is unvalidated here: test pooling mode, sign-in, spend writes and migrations with their [advisory lock](#) before adopting it.

Startup/liveness probe `/healthz`; readiness probes `/readyz` (infra/azure-private/lib/AppDefinition.psm1:69-71). The 300-second grace keeps ready replicas in rotation through short failovers. New sign-ins fail; existing tokens validate locally. Spend enforcement fails open by default; `enforcement.fail_closed_on_error: true` blocks inference even while ready ([outage behavior](#)). Spend limits add database work per inference request and durable counters, caps and audit tables. Back them up and test restores ([store](#); [Postgres](#)).

Foundry quota

The [Claude quota reference](#) defines subscription pools per model/version and deployment type: Global Standard across all regions; Data Zone Standard within each data zone. Another account/region in the same pool does **not** multiply quota. RPM, uncached input tokens per minute (ITPM) and output tokens per minute (OTPM) are separate limits. Cache writes count toward ITPM; cache reads do not.

These published defaults apply to Claude Sonnet 5 and Opus 5, for Global Standard and US Data Zone Standard:

Subscription offer	RPM	ITPM	OTPM
Enterprise / MCA-E	10,000	10,000,000	2,000,000
Pay-as-you-go	40	40,000	8,000

Assumption: all 10,000 requests/minute target one model, averaging 2,000 uncached input and 500 output tokens/request. Then $\text{ITPM} = 10,000 \times 2,000 = 20,000,000$ and $\text{OTPM} = 10,000 \times 500 = 5,000,000$, exceeding enterprise defaults. Recalculate per model from pilot usage responses; measured inference events contain no token counts (docs/UNKNOWNNS.md:63). Read

granted capacity on Foundry's **Quota** page and submit the [increase form](#); approval is not guaranteed. The deployment's grant remains to be checked (U-59, docs/UNKNOWNNS.md:70).

Foundry returns HTTP 429 at a rate limit. The gateway [fails over](#) in order on 429, 5xx, 401, 403, 404 or timeouts. Every request tries the first upstream; failures are not remembered. It waits up to one hour for Foundry to start responding; this is not load balancing or guaranteed prompt failover.

Example fragment: separate accounts with deployment mappings. The renderer currently emits a single account (scripts/admin/new-gateway-config.mjs:150-154); adopting this needs a renderer change, not an edit to generated output.

```
upstreams:
  - name: foundry-primary
    provider: foundry
    resource: example-primary
    auth: { use_azure_ad: true }
  - name: foundry-secondary
    provider: foundry
    resource: example-secondary
    auth: { use_azure_ad: true }
auto_include_built_in_models: false
models:
  - id: claude-sonnet-5
    upstream_model: { foundry-primary: claude-sonnet-5, foundry-secondary: claude-sonnet-5 }
  - id: claude-opus-5
    upstream_model: { foundry-primary: claude-opus-5, foundry-secondary: claude-opus-5 }
```

Mappings use upstream name ([models](#)). Each account needs private endpoint/DNS, public access and key auth disabled, and Cognitive Services User for the gateway's user-assigned identity ([Foundry auth](#); docs/adr/0005-network-restricted-deployment.md:70-72). Verify quota before relying on fallback. Sonnet uses Global Standard here (processing in any Azure region); Opus uses Data Zone Standard (processing within its zone). Preserve this on fallback; private endpoints do not change [processing residency](#).

Identity

Assign Gateway.Standard/Gateway.Premium to developer groups; Entra emits [roles](#). The gateway uses groups_claim: roles (config/gateway.azure-private.yaml:21-23). Group assignment requires Entra ID P1/P2 and excludes nested-group members ([assignment requirements](#)). Using app roles avoids relying on a groups claim that is omitted above 200 group memberships in JWTs ([group overage](#)). Per-role policies restrict models (config/gateway.azure-private.yaml:54-65); deployment.desktop: true enables Desktop (config/gateway-admin.azure-private.json:41).

Retain offline_access and test silent refresh. Sessions last 1 hour; deprovisioning takes effect at refresh/expiry, not immediately. There is no individual gateway-token revocation. Rotate signing keys in stages: accept old and new, sign with new, wait the session lifetime plus a margin, then remove old. Outright replacement signs everyone out ([session](#); [JWT rotation](#)).

Network, DNS and certificate

Developers need ExpressRoute/VPN routes to the environment's static private IP. App ingress is external to the **internal environment**, not the internet ([ingress visibility](#)). Conditionally forward the gateway zone to an [Azure DNS Private Resolver inbound endpoint](#). Link the private zone to its VNet. /login requires [private-only gateway resolution](#).

listen.trusted_proxies names the ranges a workload profile environment reserves, where the Container Apps ingress connects from, while access_control.allow_cidrs names developer networks from deployment.developerCidrs (config/gateway.azure-private.yaml:12-14; config/gateway.azure-private.yaml:67-68; config/gateway-admin.azure-private.json:35-37). Developer subnets are never trusted as proxies (docs/adr/0005-network-restricted-deployment.md:75-87).

Bind a custom domain to a TLS certificate issued by the organization's CA ([Container Apps certificates](#)); install the CA trust chain on clients. The change to the custom origin also sets GATEWAY_PUBLIC_URL and the Entra redirect URI

<origin>/oauth/callback to it; Deploy-Gateway.ps1 derives both from the default host name (infra/azure-private/lib/AppDefinition.psm1:16; infra/azure-private/lib/Steps.App.psm1:44). The CLI pins the leaf certificate per hostname. Publish its SHA-256 fingerprint and plan rotations: each prompts developers again. Put the hostname in NO_PROXY to avoid proxy bypass of pin checks ([connect developers](#)). Measured on 2026-09-29: the internal default domain served a publicly trusted, platform-managed certificate from Microsoft TLS G2 RSA CA OCSP 02, valid 2026-09-28 to 2027-01-06 (100 days); the platform controls rotation (U-56, docs/UNKNOWNNS.md:67).

Rollout waves

Load test first, then pilot all clients and corporate/VPN paths against real Foundry. Size waves from sign-in concentration and inference demand, not invitation count ([large rollouts](#)). Pause expansion when queueing, failures or quota use exceed the pilot's agreed operating bounds.

Component	Signal to watch	Source
Sign-in	sign-in refused, client_ip, rate-limited attempts; Claude Code 2.1.274+ displays The gateway is limiting sign-in attempts right now	Large rollouts
Gateway	client requests are open, latency and drain window over after; replica count, CPU, memory and restarts	Concurrency , upgrades , metrics
PostgreSQL	active_connections includes active and idle connections; compare with usable capacity, failed connections, CPU and storage	PostgreSQL metrics
Foundry	429 status in inference events; operational logs for upstream 429s hidden by successful fallback; RPM, ITPM and OTPM	Gateway logs , Claude quota
Identity	auth.denied, session.mint and session.refresh outcomes	Gateway logs
Network and TLS	Private DNS answers, connection failures and unexpected fingerprint prompts	Prerequisites , certificate pinning

Related content

- Gateway overview (docs/learn/overview.md)
- Configure models, roles and developer access (docs/learn/how-to-admin-configure.md)
- Run the inference tests (docs/learn/how-to-test-inference.md)
- Script reference (docs/learn/reference-scripts.md)
- [Deploy a network-restricted gateway](#)
- [Connect Claude Code, VS Code and Claude Desktop](#)